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ECommerce Web System

Submitted By:

Wajeeha Batool

Supervisor:

Dr Umar Daraz, Designation

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PROJECT APPROVAL FORM

CERTIFICATE OF COMPLETION

This is to certify that the following student

Student Name: Wajeeha Batool

Roll Number: BSCS-E1-018

have successfully completed their final year project titled

ECommerce Web System

in the partial fulfillment for the requirements of the Degree of Bachelor of Computer Science & Information Technology during the academic session 20xx-20xx.

Signatures:

Signatures:

Dr. Umar Daraz

Designation

Department of Computer Science

University of Sahiwal

Name of Chairman

Chairman

Department of Computer Science

University of Sahiwal

Declaration

We hereby declare that the work presented in this report is our own effort and has not been copied from any source. This report has not been submitted anywhere else for the award of any degree.

Wajeeha Batool

Supervisor's Certificate

It is certified that this thesis titled “**Ecommerce Web System**” is the result of the original work carried out by **Wajeeha Batool**, under my supervision. The work is up to the mark and is submitted in partial fulfillment of the requirements for the degree of BSCS.

Supervisor Name & Signature

Designation

Department of Computer Science

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ABSTRACT

Electronic Commerce (E-Commerce) is the process of conducting business transactions through computer networks, primarily the internet. It allows users to buy or sell products and services from the comfort of their homes, eliminating the need for physical effort involved in traditional commerce. With just a few clicks, users can browse products, compare prices, and place orders, significantly saving time and effort.

Since its emergence in the early 1990s, e-commerce has revolutionized the global business landscape. However, one of the major challenges hindering its growth, especially in developing countries, is the issue of **security**. Although significant advancements have been made in this area, ensuring safe and secure transactions continues to be a critical concern.

This project — an **Online E-Commerce Portal** — is designed to manage key elements of a digital shopping experience, including **Products, Customers, Categories, Shipping, and Payments**. The system enables the administrator to oversee and control all operations via a centralized interface. Users, such as buyers and sellers, can interact with the platform to view products, place orders, and track shipping in a seamless and organized manner.

The primary objective of this web application is to **automate and streamline** the e-commerce workflow by reducing manual work and enhancing the accessibility of global markets for local businesses. By integrating both backend and frontend functionalities, the platform provides a complete and user-friendly e-commerce solution tailored for developing environments.

Table of Contents

- Abstract

Chapter 1: Introduction

- Background
- Problem Statement
- Objectives
- Scope of the Project
- Methodology/Approach
- Report Organization

Chapter 2: Literature Review

- Existing Systems
- Comparative Analysis
- Gaps Identified
- Research Contributions

Chapter 3: System Analysis

- Requirements Gathering
- Functional Requirements
- Non-Functional Requirements
- Feasibility Study (Technical, Economic, Operational)
- Use Case Diagrams / DFDs

Chapter 4: System Design

- Architecture Diagram
- ER Diagram (if applicable)
- Database Design
- Interface Design
- Class Diagrams (for OOP-based systems)
- Tools & Technologies

Chapter 5: Implementation

- Frontend
- Backend
- Database Integration
- Screenshots of the Working System
- APIs/Modules Used

Chapter 6: Testing & Evaluation

- Testing Strategy (Unit, Integration, System)
- Test Cases
- Results & Observations
- Performance Evaluation
- User Feedback (if any)

Chapter 7: Conclusion & Future Work

- Summary of Achievements
- Limitations
- Proposed Enhancements

References

Chapter 1: Introduction

INTRODUCTION

Customer get many benefits via online shopping this helps e-commerce companies to build long lasting and profitable relationship with their customers. For making strong relationship with these users, it is very important to focus on the customer as a whole and making sense of a flood of real time information that goes well beyond demographics or shopping behavior. There are two entities who will have the access to the system. One is the admin and another one will be the registered user. Admin can add product details, view all the order details and can also view the sales of the products. User need to register with basic registration details to generate a valid username and password. After the user logins, it can view all the products that are recommended on the homepage compiled by the system based on user's information. From the recommended products, the user can even further view its details and then if interested to buy, the system gives add to cart option for purchasing the product. The system even has an AI bot with the help of which the user can get answers to queries like features, warranty, price etc. details of the products. This AI Bot even converts text to speech. After selecting the product, user can do payment for the particular product online. Users can view their order history of their purchased product.

Problem Statement

Currently the customer is visiting the shop to buy the products.

- The main problem for the customer is that it is a time-consuming task to visit the shop to buy the products.
- Therefore, customer needs to visit the website, browse for the products of his wish list and place the order accordingly.
- **LIMITATION OF CURRENT SYSTEM IN USE:** Wastage of time in visiting the shop. More efforts to carry the products back to their respective places, if lot of products have to be buoyed. Difficult to recollect the products needed. More human efforts as all calculation are manually done.

Objective of the Study

Generally, the objective of the study is to develop a buy and sell website to assist the marketing process of Momal, Pakistan Occidental.

Specifically, the study aimed:

1. To design a web-page that enhances marketing through:
 - i Online product advertising;
 - ii Online buying and selling.
2. To design a web-page that enable customers to:
 - i Search product online
 - ii View products description
3. To create a web portal where consumer can purchase and order products.

Scope of the Project

Project Objective and Scope of Project:

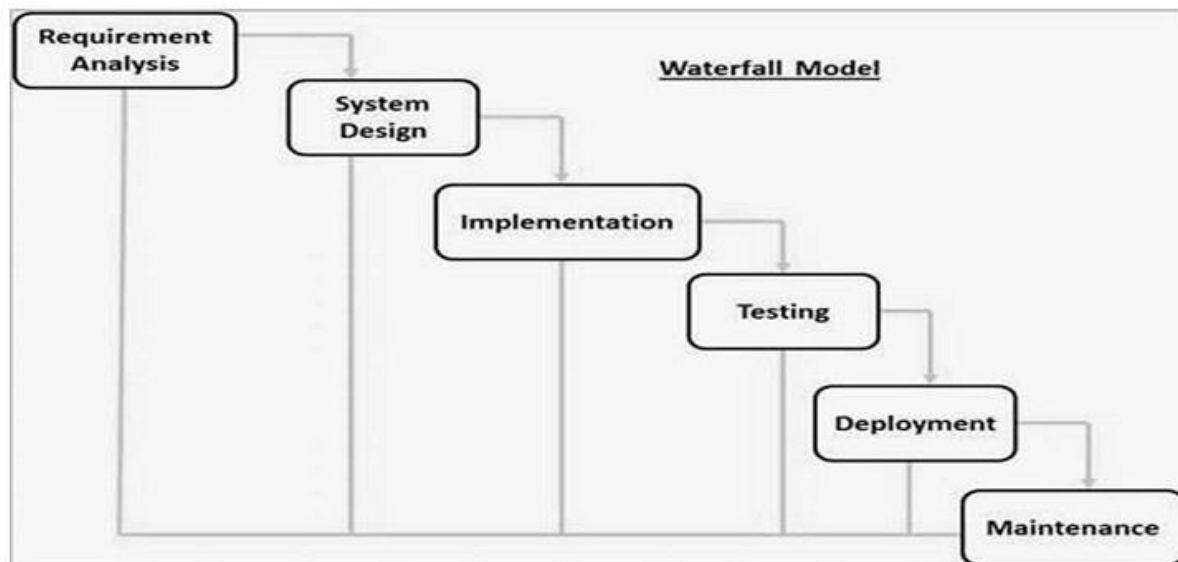
Project Objective The main objective behind making this website is increase the business, smarter and better.

The website meets the following objective that Online Shopping Store needs:

- Faster bill and receipt generation (Admin side).
- Cutting time in gathering details of product manually.
- Online bills hence optimizing the use of page (Admin Side). Scope of Project At presently this website is support for just placing order and and bill register. Further enhancement of the software will have the following features:
 - Alerts to customers of Online Shopping Store when order is generated via e-mail.
 - A website will allow customer to pay the bill through online payment.

Methodology Development Model

The sequential phases in Waterfall model are – Requirement Gathering and analysis – All possible requirements of the system to be developed are captured in this phase and documented in a requirement specification document.



- **System Design** – The requirement specifications from first phase are studied in this phase and the system design is prepared. This system design helps in specifying hardware and system requirements and helps in defining the overall system architecture.
- **Implementation** – With inputs from the system design, the system is first developed in small programs called units, which are integrated in the next phase. Each unit is developed and tested for its functionality, which is referred to as Unit Testing.
- **Integration and Testing** – All the units developed in the implementation phase are integrated into a system after testing of each unit. Post integration the entire system is tested for any faults and failures.
- **Deployment of system** – Once the functional and non-functional testing is done; the product is deployed in the customer environment or released into the market.
- **Maintenance** – There are some issues which come up in the client environment. To fix those issues, patches are released. Also to enhance the product some better versions are released. Maintenance is done to deliver these changes in the customer environment.

INTEGRATION TESTING

Integration testing (sometimes called integration and testing, abbreviated I&T) is the phase in software testing in which individual software modules are combined and tested as a group. It occurs after unit testing and before validation testing. Integration testing takes as its input modules that have been unit tested, groups them in larger aggregates, applies tests defined in an integration test plan to those aggregates, and delivers as its output the integrated system ready for system testing.

PURPOSE

The purpose of integration testing is to verify functional, performance, and reliability requirements placed on major design items. These "design items", i.e., assemblages (or groups of units), are exercised through their interfaces using black-box testing, success and error cases being simulated via appropriate parameter and data inputs. Simulated usage of shared data areas and inter-process communication is tested and individual subsystems are exercised through their input interface. Test cases are constructed to test whether all the components within assemblages interact correctly, for example across procedure calls or process activations, and this is done after testing individual modules, i.e., unit testing. The overall idea is a "building block" approach, in which verified assemblages are added to a verified base which is then used to support the integration testing of further assemblages. Software integration testing is performed according to the software development life cycle (SDLC) after module and functional tests. The cross dependencies for software integration testing are: schedule for integration testing, strategy and selection of the tools used for integration, define the cyclomatic complexity of the software and software architecture, reusability of modules and life-cycle and versioning management. Some different types of integration testing are big-bang, top-down, and bottom-up, mixed (sandwich) and risky-hardest. Other Integration Patterns [2] are: collaboration integration, backbone integration, layer integration, client-server integration, distributed services integration and high-frequency integration.

Big Bang

In the big-bang approach, most of the developed modules are coupled together to form a complete software system or major part of the system and then used for integration testing. This method is very effective for saving time in the integration testing process. However, if the test cases and their results are not recorded properly, the entire integration process will be more complicated and may prevent the testing team from achieving the goal of integration testing. A type of big-bang integration testing is called "usage model testing" which can be used in both software and hardware integration testing. The basis behind this type of integration testing is to run user-like workloads in integrated user-like environments. In doing the testing in this manner, the environment is proofed, while the individual components are proofed indirectly through their use. Usage Model testing takes an optimistic approach to testing, because it expects to have few problems with the individual components. The strategy relies heavily on the component developers to do the isolated unit testing for their

product. The goal of the strategy is to avoid redoing the testing done by the developers, and instead flesh-out problems caused by the interaction of the components in the environment.

Top-down And Bottom-up

Bottom-up testing is an approach to integrated testing where the lowest level components are tested first, then used to facilitate the testing of higher-level components. The process is repeated until the component at the top of the hierarchy is tested. All the bottom or low-level modules, procedures or functions are integrated and then tested. After the integration testing of lower-level integrated modules, the next level of modules will be formed and can be used for integration testing. This approach is helpful only when all or most of the modules of the same development level are ready. This method also helps to determine the levels of software developed and makes it easier to report testing progress in the form of a percentage. Top-down testing is an approach to integrated testing where the top integrated modules are tested and the branch of the module is tested step by step until the end of the related module. Sandwich testing is an approach to combine top-down testing with bottom-up testing.

ORGANISATION OF THE REPORT

INTRODUCTION

This section includes the overall view of the project i.e. the basic problem definition and the general overview of the problem which describes the problem in layman terms. It also specifies the software used and the proposed solution strategy.

SOFTWARE REQUIREMENTS SPECIFICATION

This section includes the Software and hardware requirements for the smooth running of the application.

DESIGN & PLANNING

This section consists of the Software Development Life Cycle model. It also contains technical diagrams like the Data Flow Diagram and the Entity Relationship diagram.

IMPLEMENTATION DETAILS

This section describes the different technologies used for the entire development process of the Front-end as well as the Back-end development of the application.

RESULTS AND DISCUSSION

This section has screenshots of all the implementation i.e. user interface and their description.

SUMMARY AND CONCLUSION

This section has screenshots of all the implementation i.e. user interface and their description.

Chapter 2: Literature Review

REVIEW OF RELATED LITERATURE

According to Basu and Muylle (2007) companies can gain two fundamental types of benefits from e-Business. There are generally described as Value Creation or Value Enhancement for one or more of a company's stakeholder groups, and Lower cost of providing goods and services to the market place under Value Creation include Improvement in internal and external communication through effective e-marketing. Increment of sales through an e-commerce website integrated with a back-office systems and Improvement in supplier relations and productivity through collaborative workspaces. And examples under Lower cost are reduction in communication and travel cost using online meeting tools, shared workspaces and benefits license free open-source alternatives to proprietary software.

In this support Xiao and Benbasat (2007) have also explain that concept of e-commerce is quite wide as it includes business to business

(B2B), Business to customer (B2C) and Customer to customer (C2C) transaction. There are different activities and function that can be undertaken online with the help of e-commerce.

Along with this, they also elaborated that in addition to financial friction, e-commerce measures are also used for the purpose of accomplishing some other important commercial transaction such as exchange of important information through electronic data interchange (EDI) in B2B business transactions and marketing of product through

internet.

There are a number of different reasons behind increasing popularity e-commerce related activities undertaken by individual and business organizations. In this context, Kim, Ferrin and Rao (2008) have elaborated that the convenience and easiness are of major consideration related with the phenomena of E-commerce. In comparison to traditional business transaction, virtual commercial transactions are quite easy and effective be carried out. It takes lesser amount of time and efforts from the side of customers.

Kim et al., (2008) have also stated that e-commerce is a new way of undertaking business transactions which are getting popularity day by day. They also illustrated that e-commerce in terms of cyber transaction in which there is a virtual transaction of money takes place between two different entities. In the views of the author, cyber transaction can be considered as the measure in which, both customer and organization interacts with different virtual means of communication such as email, mobile devices and telephones and accomplish their business transaction online.

Zhu and Thatcher (2010) have added further that e-commerce transactions save some significant amount of time to customers. Ecommerce has provided a way to the customer through which, he can easily undertake

shopping and other related business transaction while sitting at his home. Zhu and Thatcher have also stated that e-commerce measures also provide 24/7 facilities of undertaking business and commercial transactions, which cannot be possible in the case of traditional measures of business transaction.

Windrum and Berranger (2010) suggest that the commercial benefits of e-commerce lie in five areas. First, firms are able to expand their geographical reach. Second, important cost benefits lie in improved efficiency in procurement, production and logistics processes. Third, there is enormous scope for gaining through improved customer communications and management. Fourth, the Internet reduces barriers to entry for new market entrants and provides an opportunity for small firms to reorient their supply chain relationships to force new strategic partnerships.

Alnemr et al., (2010) have reflected that E-commerce is basically a process through which, customers as well as organizations undertake there all their commercial transaction with the help of internet. As per the author, there are number of different activities such as internet marketing, online transaction processing, electronic funds transfer (EFT), online payment of shopping bills, and electronic data interchange (EDI), that can be categorized under the E-commercial activities.

Swan and Nolan (2013) stated of what are the behavioral sciences know about trust to industrial sales situations. Trust between an industrial buyer and salesperson is defined cognitively and behaviorally. A model of this trust-building process is presented and appropriate behaviors for sales representatives who wish to build trust are discussed. The interactions between experience, general knowledge, and trust building are examined in light of possible situations a sales representative might encounter. The findings suggest that both firm and salesperson can act positively to build and maintain trust with their clients.

According to Pavlou (2014) the primary constructs for capturing consumer acceptance of E-commerce are intention to transact and on-line transaction behavior. This includes the theory of reasoned action (TRA) as applied to a technology-driven environment; technology acceptance model (TAM) variables (perceived usefulness and ease of use) are posited as key drivers of e-commerce acceptance. The practical utility of TAM stems from the fact that e-commerce is technology-driven. The proposed model integrates trust and perceived risk, which are incorporated given the implicit uncertainty of the e-commerce environment. The proposed integration of the hypothesized independent variables was justified by placing all the variables under the nomological TRA structure and proposing their interrelationships. The resulting research model is tested using data from two empirical studies. The first, exploratory study comprises three experiential scenarios with 103 students. The second, confirmatory study uses a sample of 155 on-line consumers. Both studies strongly support the e-commerce acceptance model by validating the proposed hypotheses.

Rogers (2015) social analytics is the end state result set that arises from social media integration with Business Intelligence, and this has created a new and exciting form of insights for business users, helping them

in understanding the actual movement of the s-curve, and assisting in creating blue ocean strategy in the marketplace. Social media can be defined as a collection of content created by people using accessible publishing technologies, thereby creating information in formats that are varied and drives user participation and potential drivers and insights into how people discover, read and share news, information and content, simply transforming into conversations and is the democratization of information, transforming people from content readers into information publishers. To benefit from integrating and connecting all the data, we will focus on going beyond just monitoring social media and the associated networking. Social Networks are becoming a mainstream part of our lives. People of all ages and backgrounds are using social networks to express opinions, share information, follow brands, and consume content. Each network provides a unique value proposition to its users and fills specific needs for interaction.

According to Heuer et al., (2015) shopping online as an alternative to in-store shopping can make much of traditional retail less attractive. There exists an obvious relationship between e-commerce and the demand for commercial real estate that accommodates physical retail shops. Substantial evidence supports the idea that the demand for physical shops has been affected by the rapid development of ecommerce in China. In order to better understand the sustainability of the commercial real estate market in a context of rapidly developing ecommerce in China, this study seeks to clarify the relationship between ecommerce and the demand for commercial real estate. Methods of inductive and comparative analysis are adopted. The findings show that from 2009 to 2013, the slowing growth rate of commercial property sales and the accelerating vacancy rate of commercial properties are closely related to the dramatic growth of e-commerce. We also found that different forms of commercial properties—supermarkets, department stores and shopping centers—are differently affected by the rise of ecommerce. However, due to the respective advantages and disadvantages associated with online and offline shopping, physical stores will not be completely eclipsed by virtual retail shopping.

According to Kemper et al., (2015) stated that when developing pricing strategies, it is highly important for managers to understand brand competition resulting from price promotions within a store. According to Madlenak et al., (2017) online marketing is an integral part of ecommerce nowadays and includes many different ways company's presentation such as e-mail marketing, context marketing, social media, affiliate marketing and so on. The fundamental metrics used in e-mail marketing as a delivery rate, open rate and click through the rate. The eye tracking technology was also used to analyze of internet users at mailbox.

According to Kosuke et al., (2016), the development of online shopping services is stimulated by both retailers and consumers. Compared to the brick-and-mortar type of stores, the online shopping retailers can save selling cost, such as rent and labor cost, and consumers who purchase items via online shopping site also can save time for shopping trips. In this study, they confirm how the online shopping affects retailer's selling cost and consumer's purchasing cost compared with the case of shopping at brick-and-mortar stores. Furthermore, they examine how delivery manners affect retailer's and consumer's cost, concluding that the online shopping retailers should clearly introduce the delivery charge independent from the price of the items.

EXISTING SYSTEM

This existing system of buying goods has several disadvantages. It requires lots of time to travel to the particular shop to buy the goods. It is having lots of manual work. Since everyone is leading busy life now a days, time means a lot to everyone. Also, there are expenses for travelling from house to shop. It is less user-friendly. In current system user must go to shop and order products. It is difficult to identify the required product. More over the shop from where we would like to buy something may not be open 24*7*365. Hence, we have to adjust our time with the shopkeeper's time or vendor's time. In current e commerce system user have to go shop to view the description of the product. It is unable to generate different kinds of report.

PROPOSED SYSTEM

The proposed system helps in building a website to buy, sell products or goods online using internet connection. Unlike traditional commerce that is carried out physically with effort of a person to go and get products, eCommerce has made it easier for human to reduce physical work and to save time. The basic concept of the application is to allow the customer to shop virtually using the Internet and allow customers to buy the items and articles of their desire from the store-commerce is fast gaining ground as an accepted and used business paradigm.

Comparative Analysis

The purpose of comparative analysis is to understand the differences and improvements between existing systems and the proposed system

Key Comparison Factors:

Feature	Existing Systems	Proposed System (Our E-Commerce System)
Platform	Mostly built using commercial frameworks like Magento, Shopify	Built from scratch using PHP and MySQL
User Registration	Basic registration, often with limited admin control	Full user and seller registration with admin verification
Product Search Functionality	Basic keyword-based search	Advanced search with filters by category, name, and price
Admin Control	Restricted or shared control	Full control to add, edit, delete users and products
Customization	Limited and costly	Fully customizable code, design, and features
Cost	Subscription-based; monthly/annually	Free after one-time development
Security	Industry standard protocols (HTTPS, 2FA, etc.)	Basic but extendable PHP security with validation and admin moderation
Scalability	Very scalable but needs resources	Suitable for small to medium businesses

Summary:

Existing platforms like Shopify or Magento offer very powerful features, but at a high cost and with limitations on customizability.

Our proposed PHP-based E-Commerce System is:

- Lightweight
- Cost-effective
- Fully customizable
- Designed especially for startups or local online shops

It provides a strong alternative where budget and specific requirements make commercial options less viable.

Gaps Identified

During the research and analysis of existing e-commerce platforms, several functional and user-experience related gaps were identified. These shortcomings often impact the efficiency, accessibility, and satisfaction of both sellers and buyers.

Key Gaps in Existing Systems:

1. Limited Admin Control:

- Most systems allow basic moderation, but full administrative control on product listings, user verification, and order tracking is missing or difficult to access.

2. Overly Complex User Interfaces:

- Some platforms are too advanced or confusing for small business owners or first-time users.
- Navigation and checkout processes are lengthy and not intuitive.

3. High Cost of Ownership:

- Premium features require monthly subscriptions or expensive plugins.
- Small businesses cannot afford these platforms.

4. Rigid Architecture:

- Existing platforms are not easily customizable for unique business models (e.g., custom categories, discount logic).

5. Basic Search Filters:

- Search features are often limited to keywords only, without filters like price, category, or brand.

6. Security Limitations for Small Deployments:

- Self-hosted systems often lack built-in protection for data validation, SQL injection, and spam filtering.

Our Proposed System Addresses These Gaps by:

- Giving the admin full control over all entities: products, users, orders, and inventory.
- Designing a simple, clean user interface suitable for all levels of users.
- Building the system in pure PHP, making it free, open-source, and easily modifiable.
- Including advanced search filters for smoother product discovery.
- Implementing basic PHP security techniques like input validation and role-based access control.
- Supporting custom discount strategies and flexible business rules.

Problem Analysis

Ecommerce system is a computerized, online solution to the various problems faced by the Product buyer and seller wishing to outsource their software development work to a Provider at an economical cost, thus achieving high performance, accuracy, reliability and high speed of data retrieval.

In this system, there is a registration process each for the Product buyer and seller. The Administrator of the site verifies the Provider after his registration and if satisfied, assigns him a user name and password.

Our site can be used by anyone who is searching for Products whether he/she is first time visiting our site. Our site also provides some discounted Products as same u get on any shop.

The software covers the following point while keeping in mind user's requirement-:

- Fast online access of information about various Products.
- Search Products by keywords like functional area, experience and also by initials of the Product's name.
- Administrator will maintain the database and perform all process.

Research Contributions

The development of the Online E-Commerce System contributes significantly to the field of small-scale digital commerce, especially in terms of accessibility, cost-effectiveness, and customizability. This project not only solves common problems in existing systems but also introduces innovative solutions tailored for startups and small businesses.

Major Contributions:

- 1. Custom-Built E-Commerce Platform in PHP:**
 - Unlike commercial systems, this project is developed from scratch using PHP and MySQL, making it lightweight and easy to deploy on shared hosting environments.
- 2. Full Admin Control Module:**
 - The admin has complete authority over managing users, products, categories, orders, and discount offers — which is rarely provided in low-cost systems.
- 3. Advanced Search and Filtering:**
 - Product search includes filtering options like category, keyword, and price range, improving overall user experience and shopping convenience.

4. User Verification and Role Management:

- The system includes a secure registration process for both buyers and sellers, with admin approval for seller accounts, reducing the chances of spam or fraudulent listings.

5. Dynamic Discount System:

- Admins can add smart discount schemes, apply category-wise discounts, or even enable promotional offers, giving full control over pricing strategies.

6. Database-Driven Product Management:

- All products are dynamically managed through the database with easy CRUD operations (Create, Read, Update, Delete), enabling real-time updates without code changes.

7. User-Friendly UI/UX:

- Simple and responsive design that works smoothly on both desktop and mobile browsers, allowing wider access to users with different devices.

8. Scalable Design:

- The modular structure of the system allows easy addition of new features in future, such as payment gateways, chat support, or product reviews.

Academic and Practical Impact:

- This project can serve as a starter kit for other students and developers aiming to build e-commerce solutions for local businesses.
- It also helps understand the end-to-end lifecycle of a real-world software project — from requirement analysis to deployment.

Chapter 3: System Analysis

System Analysis

The objective of the system analysis activity is to develop structured system specification for the proposed system. The structured system specification should describe what the proposed system would do; independent of the technology, which will be used to implement these requirements. The structured system specification will be used to implement these requirements. The structured system specification will be called the essential model (also known as logical model).

The essential model may itself consist of multiple models, modeling different aspect of the system. The data flow diagrams may model the data and their relationships and the state transition diagram may model time dependent behavior of the system. The essential model thus consists of the following.

- Context diagram
- Leveled data flow diagrams
- Process specification for elementary bubbles
- Data dictionary for the flow and stores on the DFDs.

Hardware Requirements

Number	Description
1	PC with 250 GB or more Hard disk.
2	PC with 2 GB RAM.
3	PC with Pentium 1 and above.

Software Requirements

Number	Description	Type
1	Operating System	Windows XP / Windows
2	Language	PHP
3	Database	MySQL
4	IDE	Visual Code
5	Browser	Google Chrome

REQUIREMENT SPECIFICATION

Functional Requirements

The software that we are making has the following functional requirement:

- Functions of Website:
- ✓ User can search various products by typing only the keyword
- ✓ Handling multiple customers and their accounts.
- ✓ User profile customization i.e. by adding his/her profile photo
 - Admin Functionality:
- ✓ Add, edit and remove Customer
- ✓ Add, edit and remove Users.
- ✓ Add, view the products.
 - User Functionality:
- ✓ View Bill.
- ✓ View products and shop for them.

Technical Requirements:

- For running website there are some specifications and only if it satisfies the specified requirement, it will get installed and works as it is designed for.
 - A technical requirement pertains to the technical aspects that your system must fulfil, such as performance-related issues, reliability issues, and availability issues. These types of requirements are often called quality of service (QoS) requirements, service- level requirements or non-functional requirements.
 - For our software also there are some specifications there are as follows:
- ✓ Minimum space required in HDD is 50GB.
 - ✓ RAM required is 2GB.
 - ✓ Screen resolution should be 1024x768 or more.
 - ✓ Internet Connection.
 - ✓ Web browser with latest version or any version that is not old than a year.
 - ✓ Pen drive or external HDD for safe backup.
 - ✓ Windows 10,8.1,7 or any higher version.
 - ✓ Works with both 32bit and 64bit Operating System.
 - ✓ XAMPP Server

The website will work properly as intended and smoothly only if the requirements are met or else the software will work but it may lag.

Software Requirement Specification:

The Online Shopping Store Stores is designed with the aim of placing an order from anywhere and anytime.

- The system is capable to handle more than one customer at a same time once it is online.
- The system will have a login that keeps track of different customers and also Admin and customer have different rights.
- Admin Functionality:
 - i Add, edit and remove Product.
 - ii Add, edit and remove Users.
 - iii View bill.
 - iv Backup and Restore data.
- User Functionality:
 - View Order Details

As per the requirement and guidelines that were given, we have formulated this document and by signing this document we accept that all the requirements are met.

FEASIBILITY STUDY

Operational Feasibility

Operational feasibility is to gain an understanding of whether the proposed system will likely solve the business problems, or take advantage of the opportunities or not. It is important to understand how the new system will fit the current day-to-day operations of the organization.

- We have also conducted an operational feasibility of our system to identify whether our system is able to satisfy the goal of developing the system.
- The other major requirement was to develop website for growth of business. This requirement was also satisfied.
- The next thing was to make a solution to get the product details
- The next thing we needed was a system that is easy to use. So, we designed an interface which is extremely easy to use and also, we have given attention to website wherein customer can access the website from computer.

Technical Feasibility:

Assessing technical feasibility is to evaluate whether the new system will perform adequately and whether an organization has ability to construct a proposed system or not. The technical assessment help answer the question such as whether the technology needed for the system exists, how difficult it will be to build, and whether the firm has enough experience using that technology.

- In case of technical feasibility, we have made a system that match accord to the requirements of the accountant.
- We have made the project in a language in which we are familiar. The entire project and website are made in PHP and MySQL Server. We have chosen these to as the project is small and also, we are familiar with the technologies.

The next area where have concentrated is the operating system architecture.

We are currently using Windows 10, we have made the website, operating system and architecture friendly.

Economic Feasibility:

Economic evaluation is a vital part of investment appraisal, dealing with factors that can be quantified, measured, and compared in monetary terms (Chen 1996). The results of an economic evaluation are considered with other aspects to make the project investment decision as the proper investment appraisal helps to ensure that the right project is undertaken in a manner that gives it the best chances of success.

- The system according to us is cost effective as this a small group which is containing little members and only two members are working on this project

Use Case Diagram:

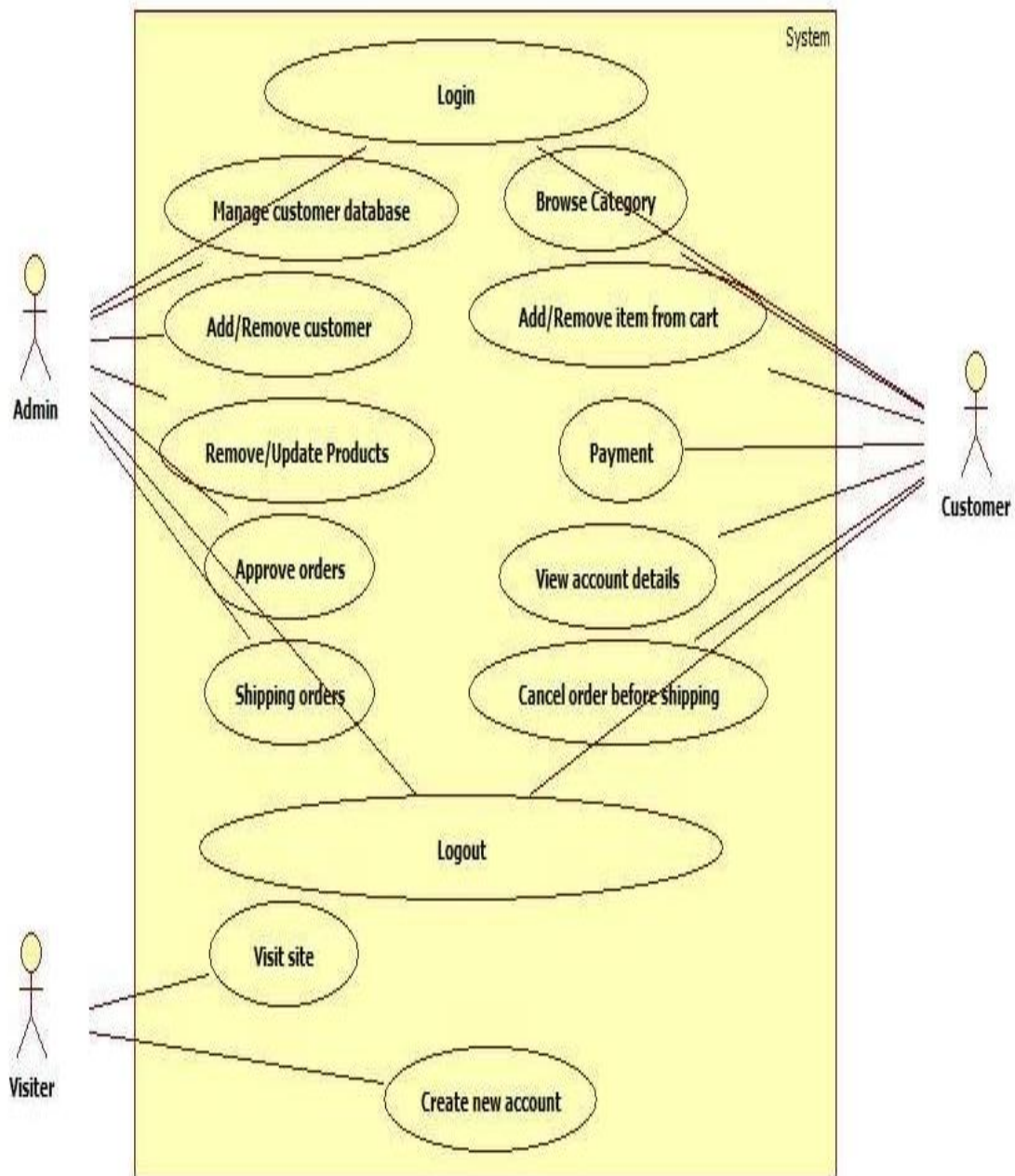


fig. Use Case Diagram

Chapter 4: System Design

System Design

System design involves transformation of the user implementation model into software design. The design specification of the proposed system consists of the following:

- Database scheme
- Structure charts
- Pseudo codes for the modules in structure charts.

E-R Diagram:

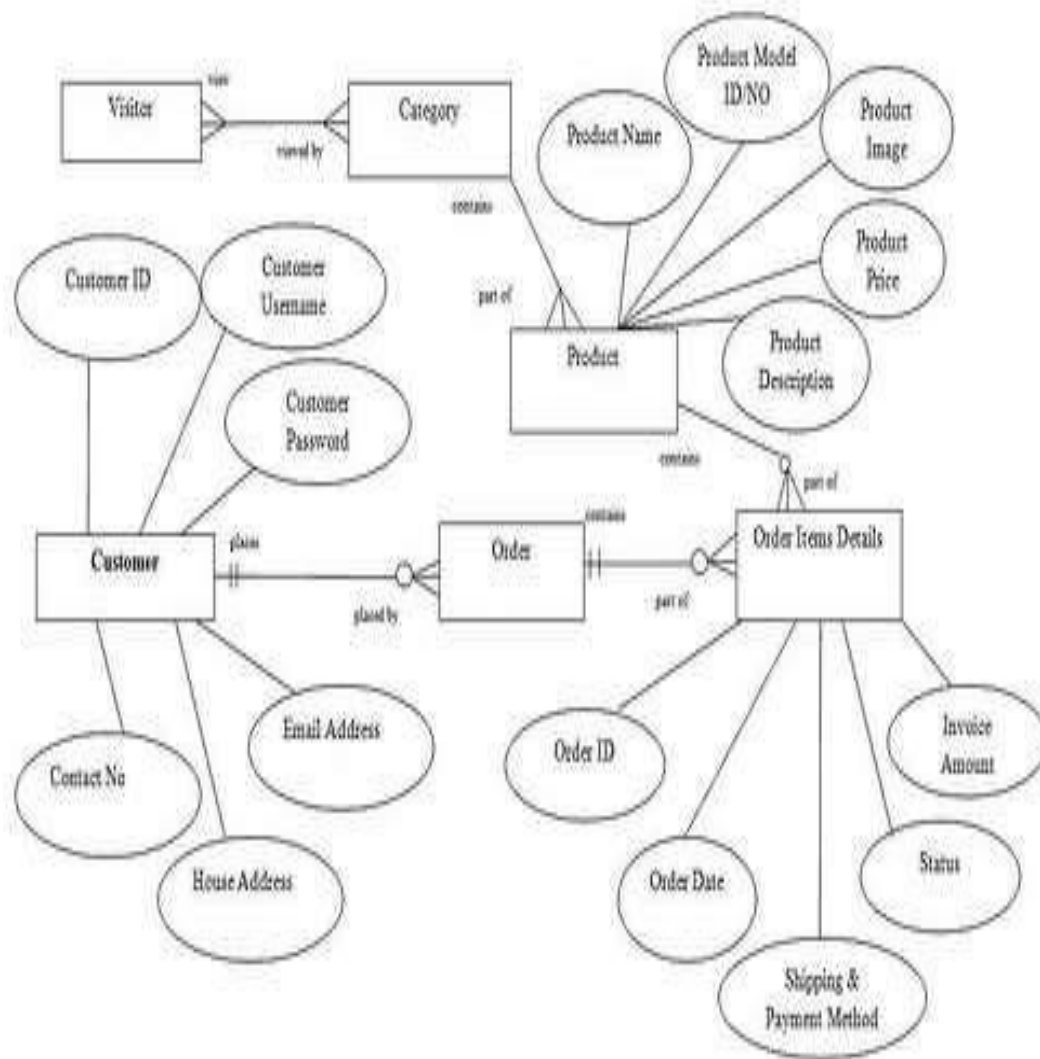


fig. E-R Diagram

List of tables with constraints:

- SQL constraints are used to specify rules for the data in a table.
- Constraints are used to limit the type of data that can go into a table. This ensures the accuracy and reliability of the data in the table. If there is any violation between the constraint and the data action, the action is aborted.
- Constraints can be column level or table level. Column level constraints apply to a column, and table level constraints apply to the whole table.

- Admin

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	user_id 	int(10)			No	None		AUTO_INCREMENT
2	user_email	varchar(255)	latin1_swedish_ci		No	None		
3	user_pass	varchar(255)	latin1_swedish_ci		No	None		

fig. Admin Table Structure

- Brand

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	brand_id 	int(100)			No	None		AUTO_INCREMENT
2	brand_title	text	latin1_swedish_ci		No	None		

fig. Brand Table Structure

- Cart

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	p_id 	int(10)			No	None		
2	ip_add	varchar(255)	latin1_swedish_ci		No	None		
3	qty	int(10)			No	None		

fig. Cart Table Structure

- Categories

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	cat_id 	int(100)			No	None		AUTO_INCREMENT
2	cat_title	text	latin1_swedish_ci		No	None		

fig. Categories Table Structure

|

- Customers

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	customer_id 	int(10)			No	None		AUTO_INCREMENT
2	customer_ip	varchar(255)	latin1_swedish_ci		No	None		
3	customer_name	text	latin1_swedish_ci		No	None		
4	customer_email	varchar(100)	latin1_swedish_ci		No	None		
5	customer_pass	varchar(100)	latin1_swedish_ci		No	None		
6	customer_country	text	latin1_swedish_ci		No	None		
7	customer_city	text	latin1_swedish_ci		No	None		
8	customer_contact	varchar(255)	latin1_swedish_ci		No	None		
9	customer_address	text	latin1_swedish_ci		No	None		
10	customer_image	text	latin1_swedish_ci		No	None		

fig. Customers Table Structure

- Orders

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	order_id 	int(100)			No	None		AUTO_INCREMENT
2	p_id 	int(100)			No	None		
3	c_id	int(100)			No	None		
4	qty	int(100)			No	None		
5	invoice_no	int(100)			No	None		
6	status	text	latin1_swedish_ci		No	None		
7	order_date	date			No	None		

fig. Orders Table Structure

- Payments

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	payment_id 	int(10)			No	None		AUTO_INCREMENT
2	invoice_no	int(10)			No	None		
3	amount	int(10)			No	None		
4	payment_mode	text	latin1_swedish_ci		No	None		
5	ref_no	int(10)			No	None		
6	code	int(10)			No	None		
7	payment_date	text	latin1_swedish_ci		No	None		

fig. Payments Table Structure

- Products

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	product_id 	int(100)			No	None		AUTO_INCREMENT
2	product_cat	int(100)			No	None		
3	product_brand	int(100)			No	None		
4	product_title	varchar(255)	latin1_swedish_ci		No	None		
5	product_price	int(100)			No	None		
6	product_desc	text	latin1_swedish_ci		No	None		
7	product_image	text	latin1_swedish_ci		No	None		
8	product_keywords	text	latin1_swedish_ci		No	None		

fig. Products Table Structure

Class Diagram:

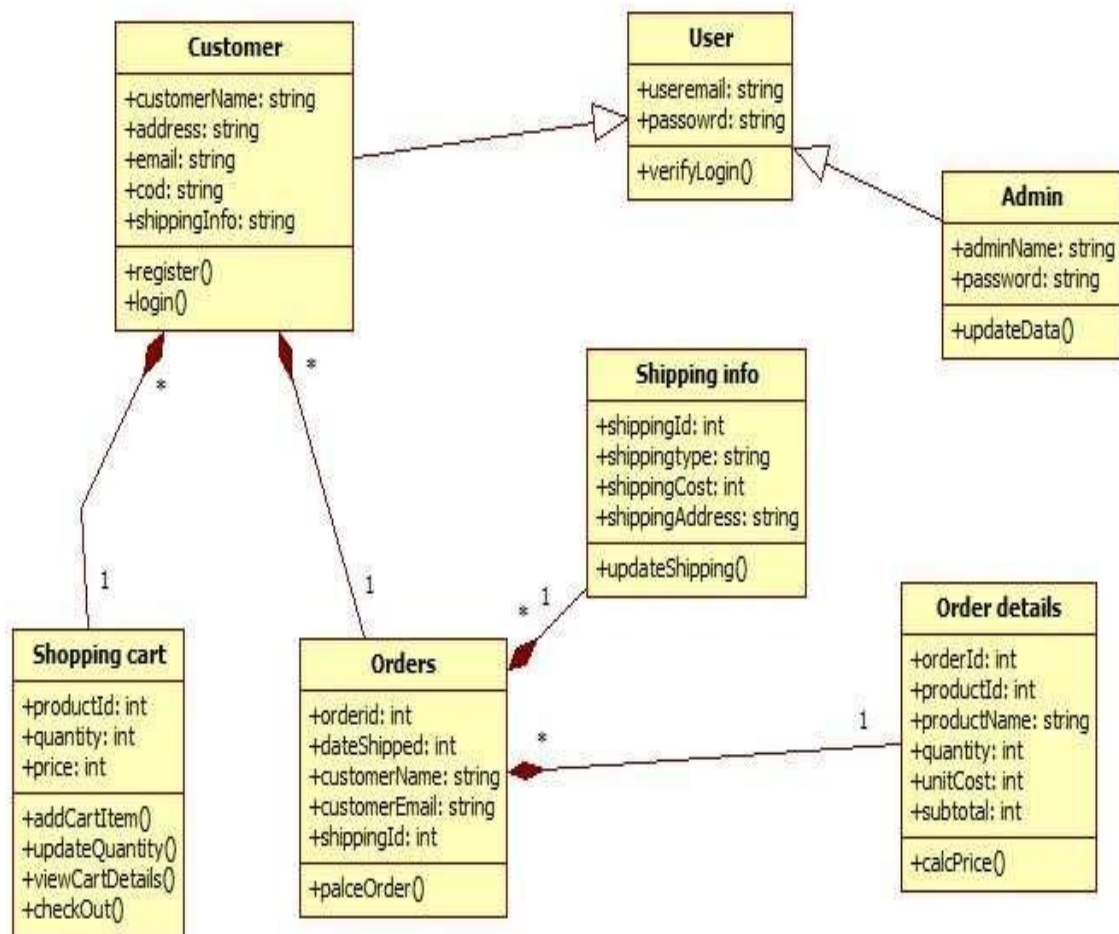


fig. Class Diagram

Activity Diagram:

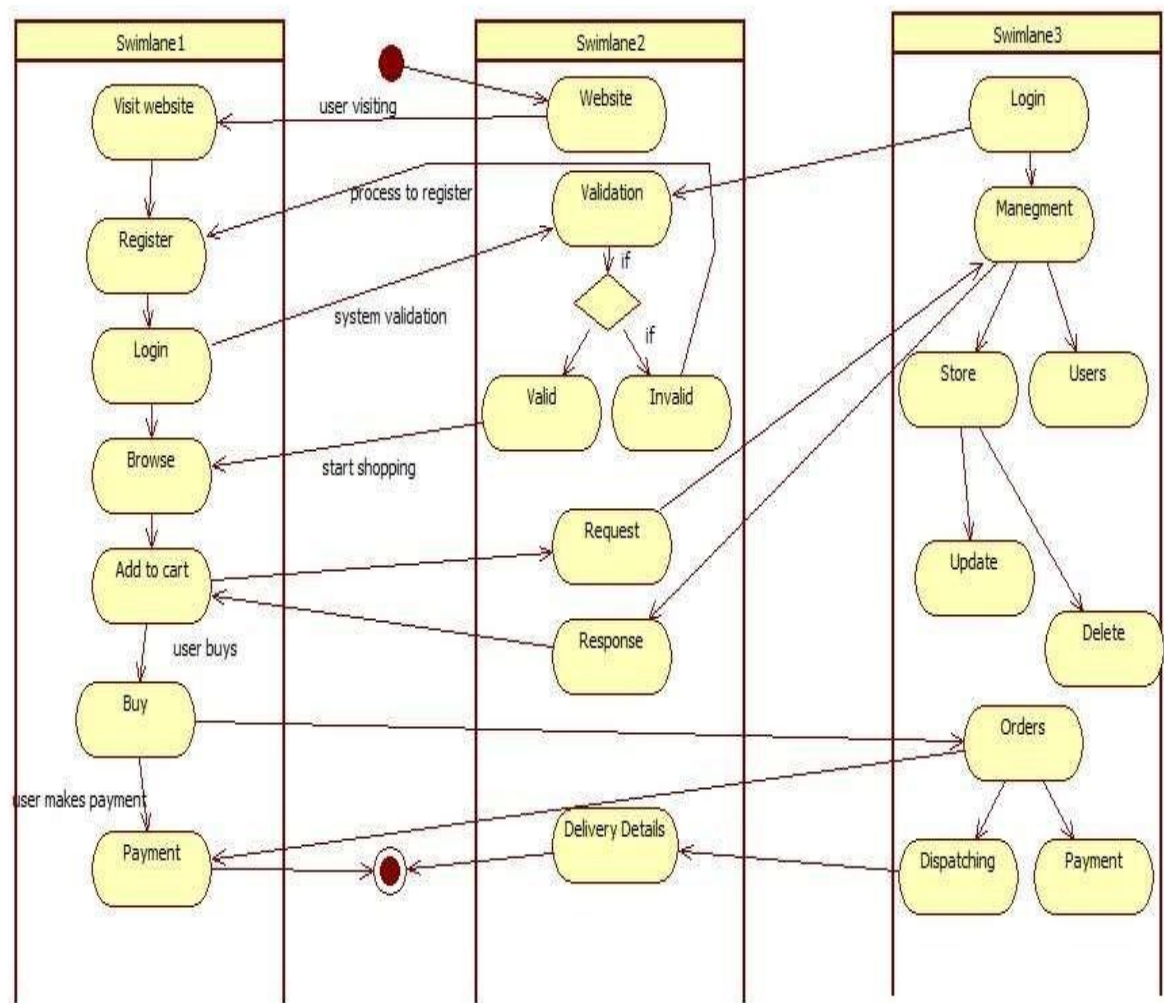


fig. Activity Diagram

Sequence Diagram:

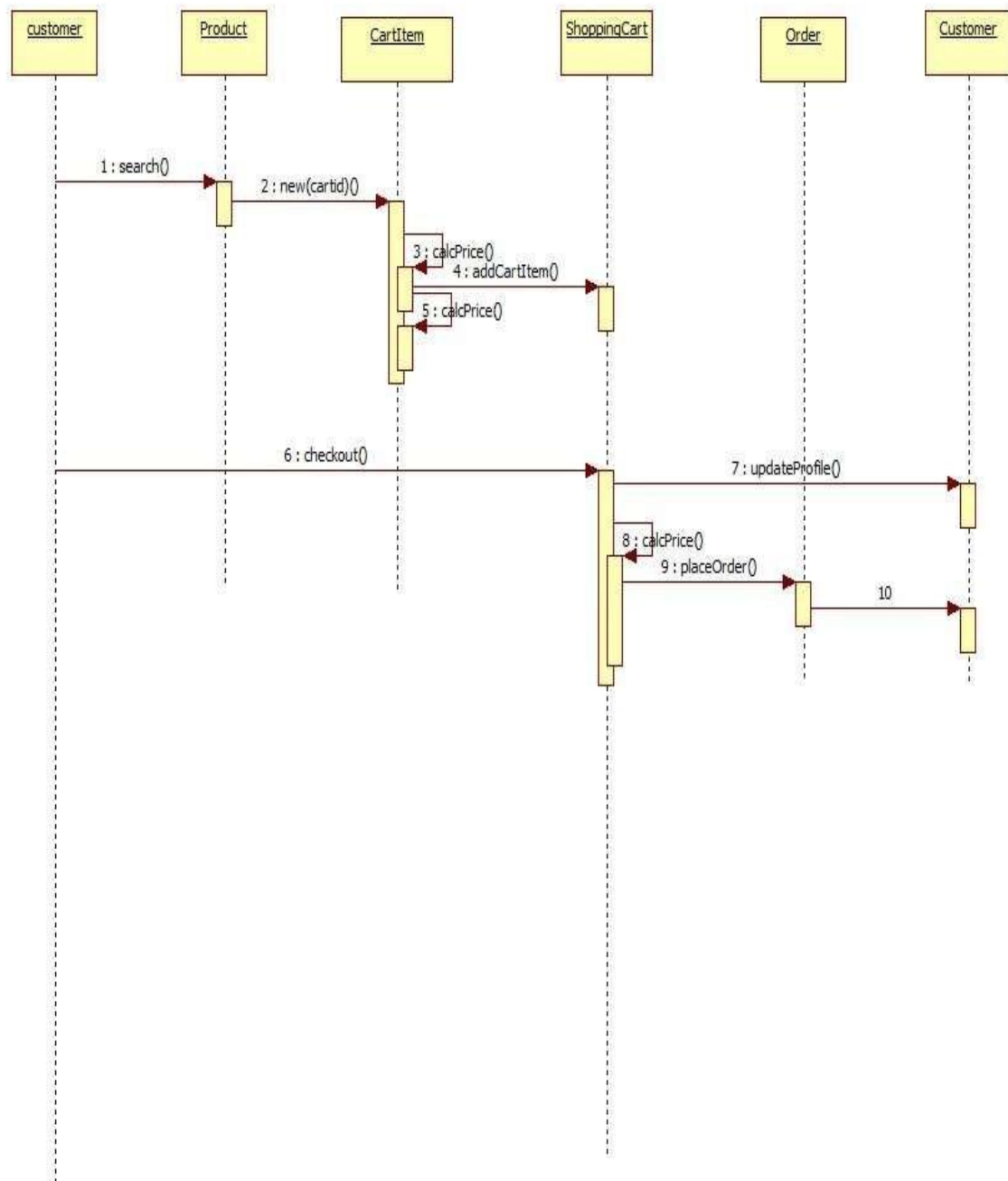


fig. Sequence Diagram

State Machine Diagram (Admin):

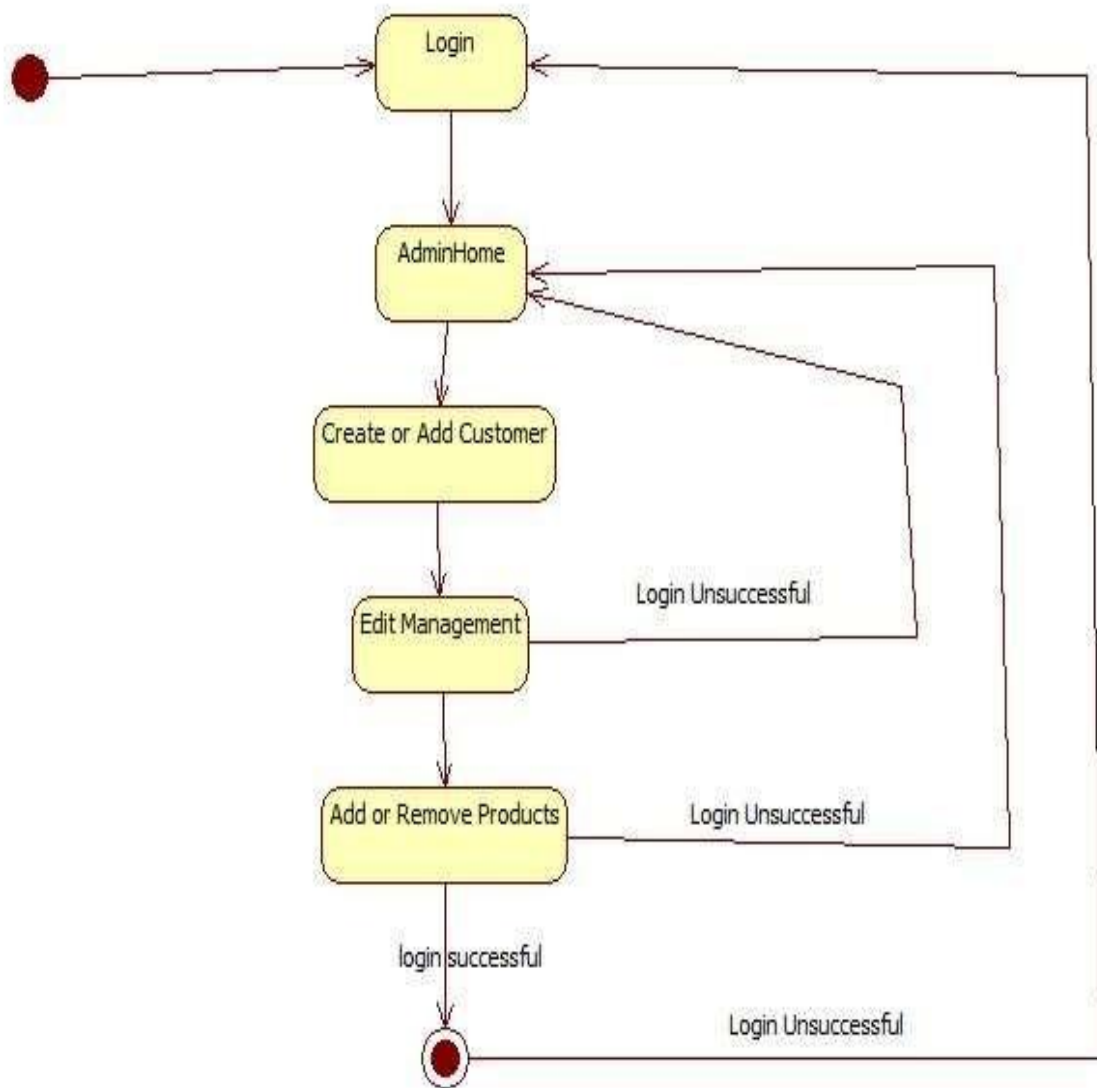


fig. State Machine Diagram (Admin)

State Machine Diagram (Customer):

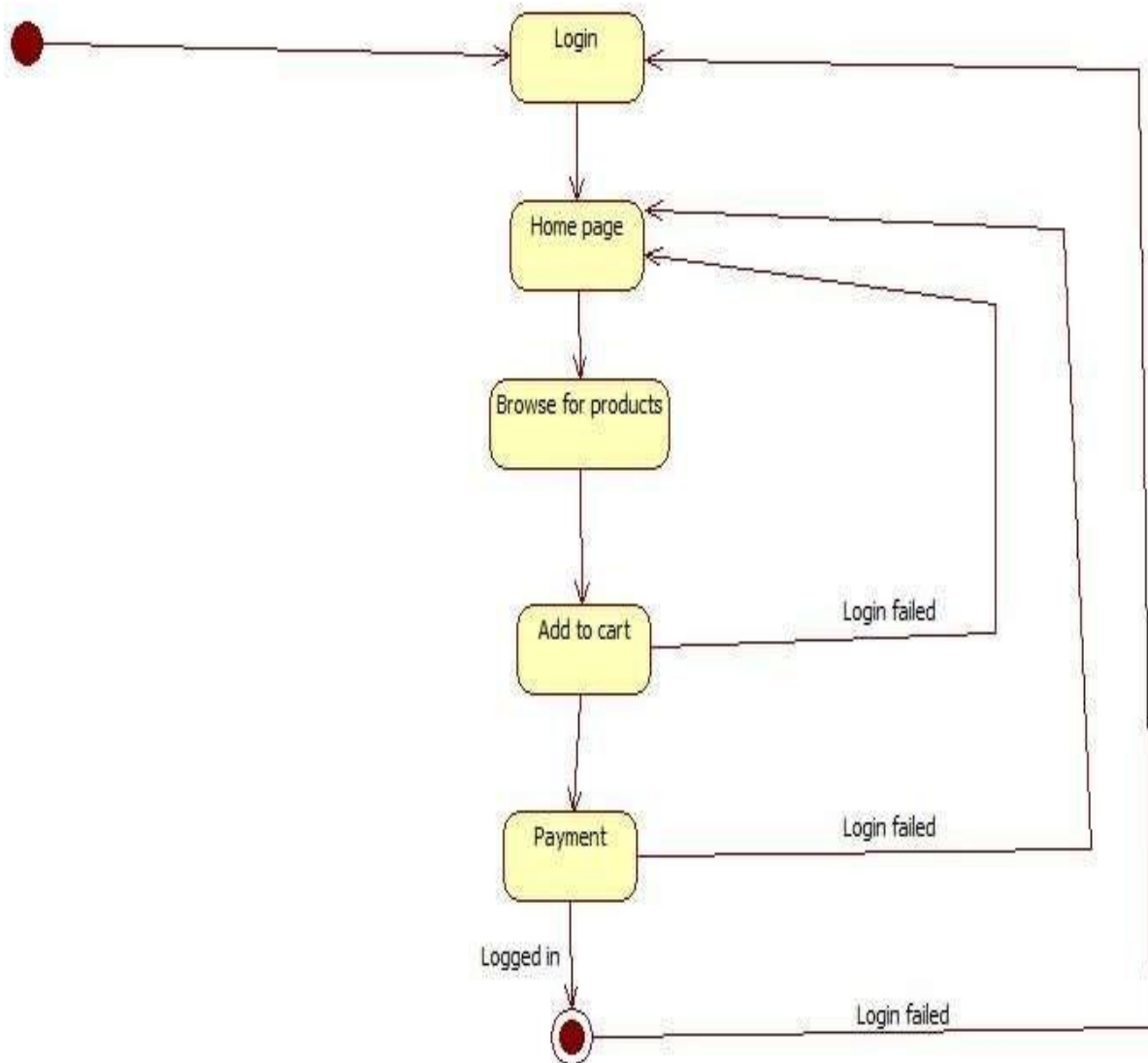


fig. State Machine Diagram (Customer)

Component Diagram:

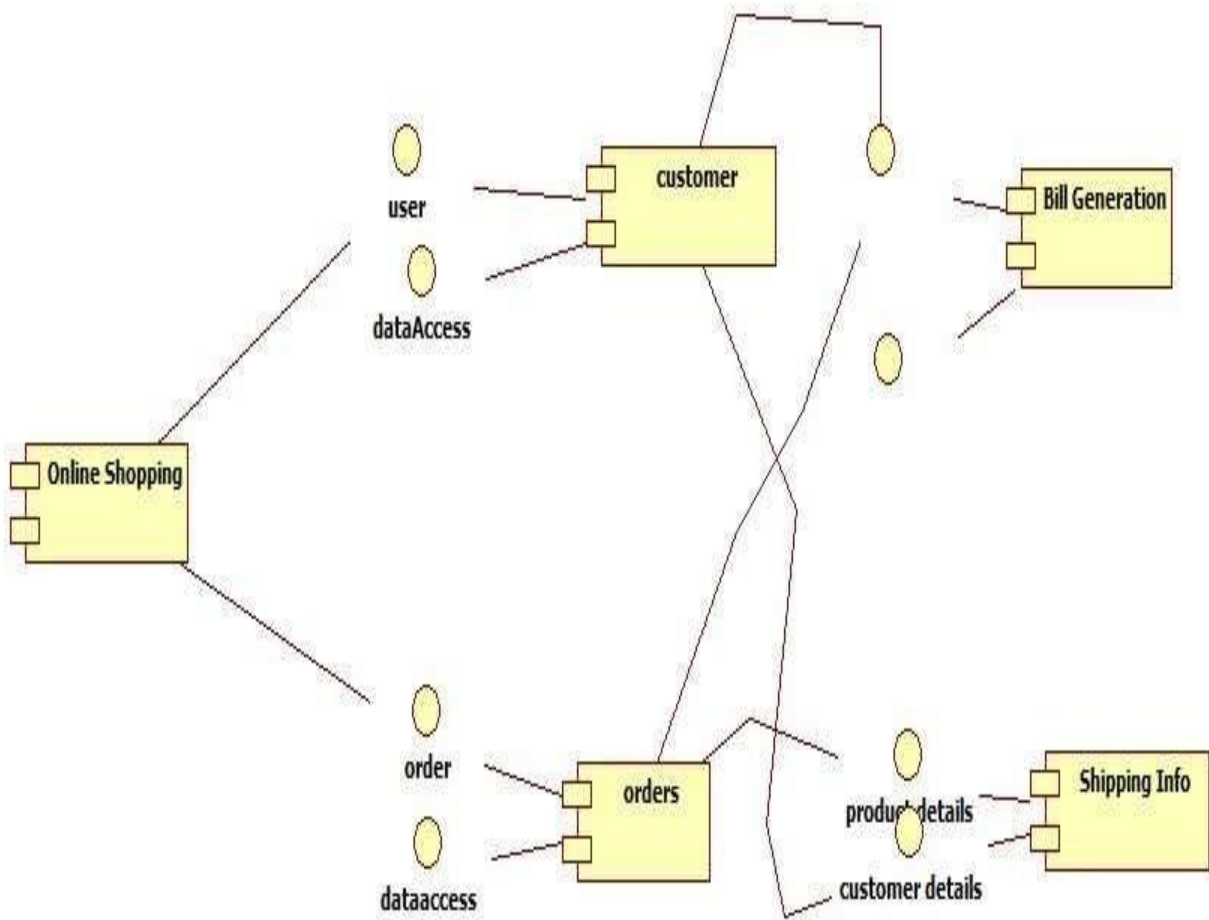


fig. Component Diagram

Deployment Chart:

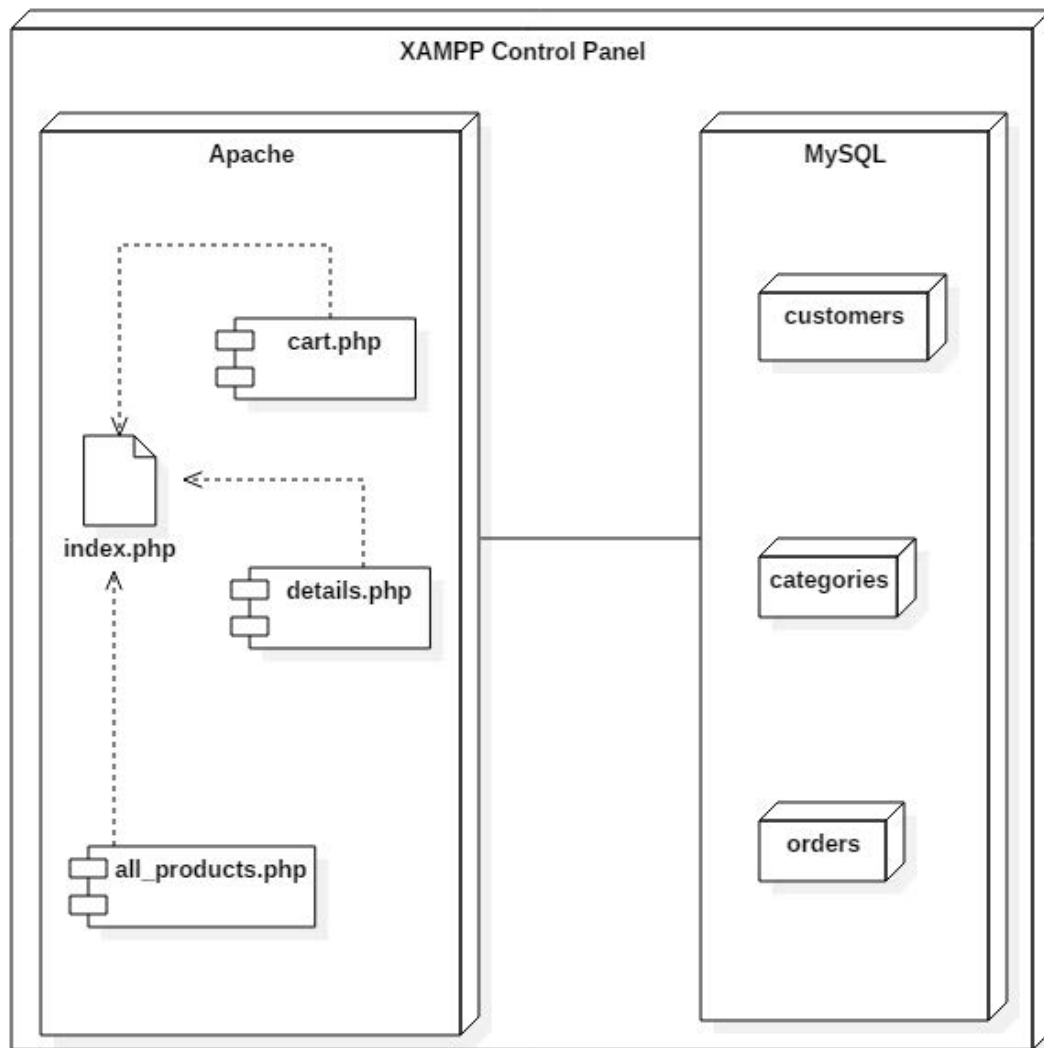


fig. Deployment Chart

Site Map:

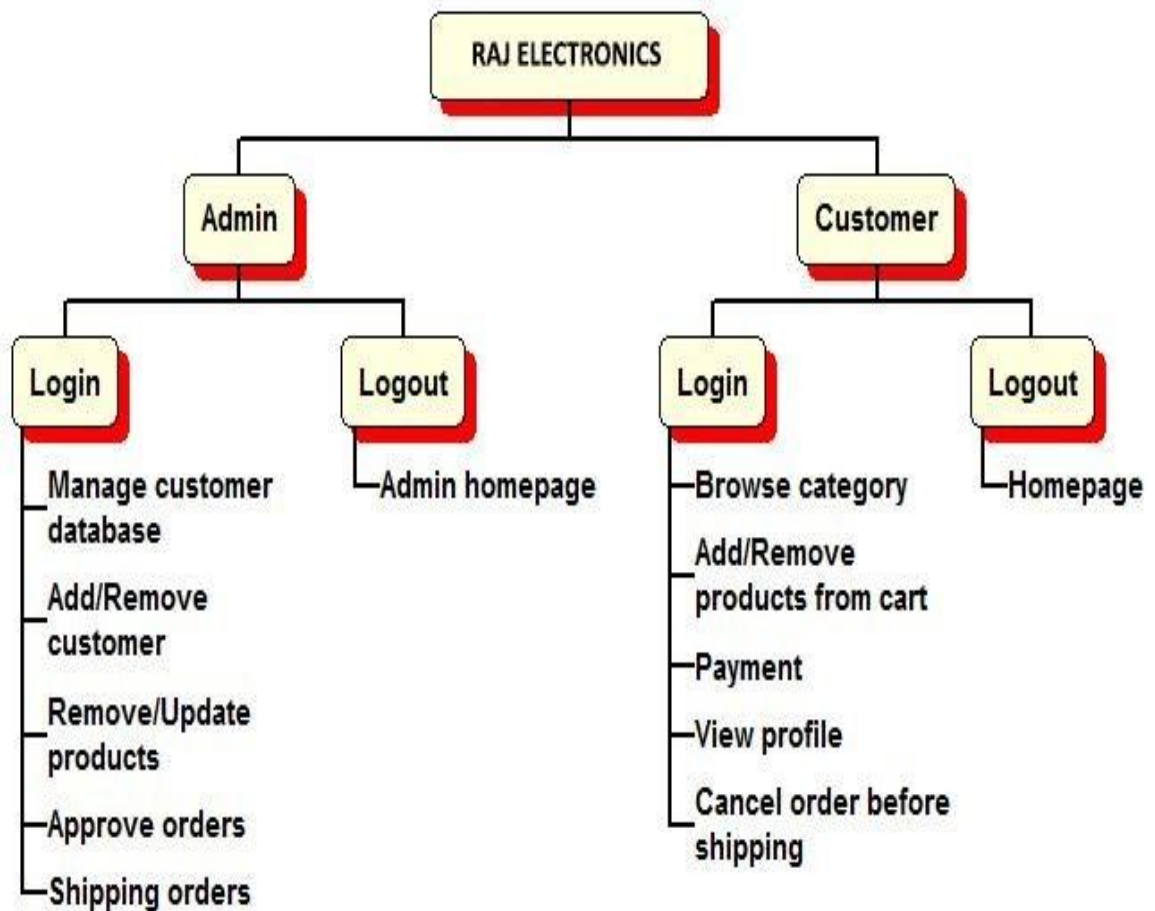


fig. Site Map

System flow chart Admin flow chart:

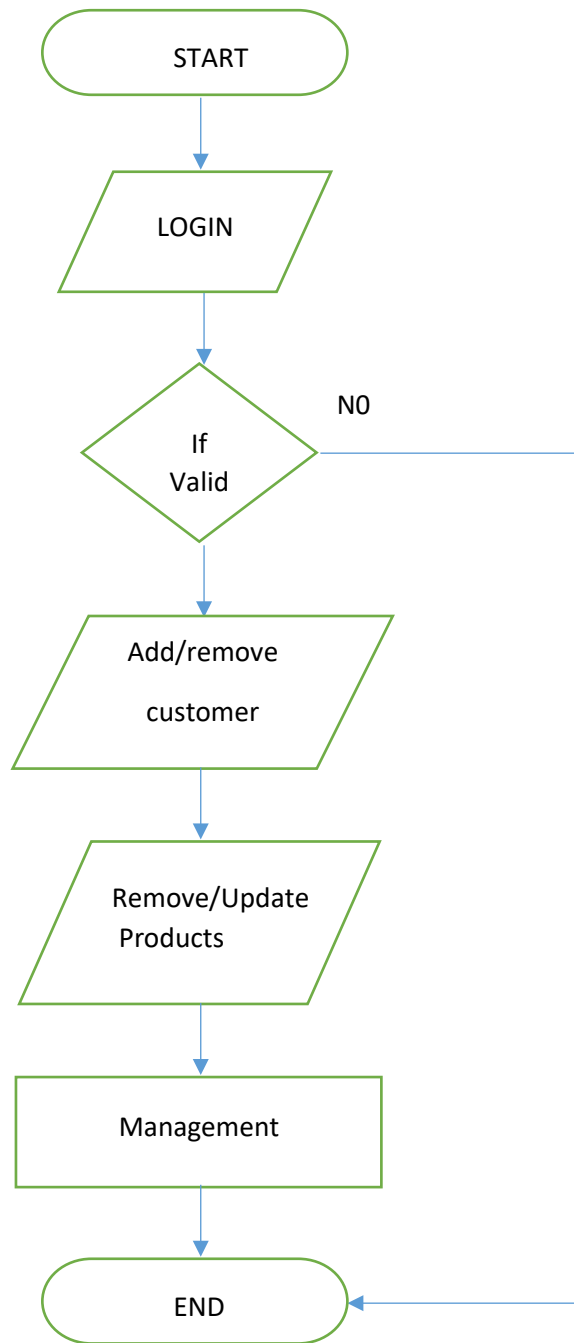


fig. Admin Flow Chart

Customer flow chart:

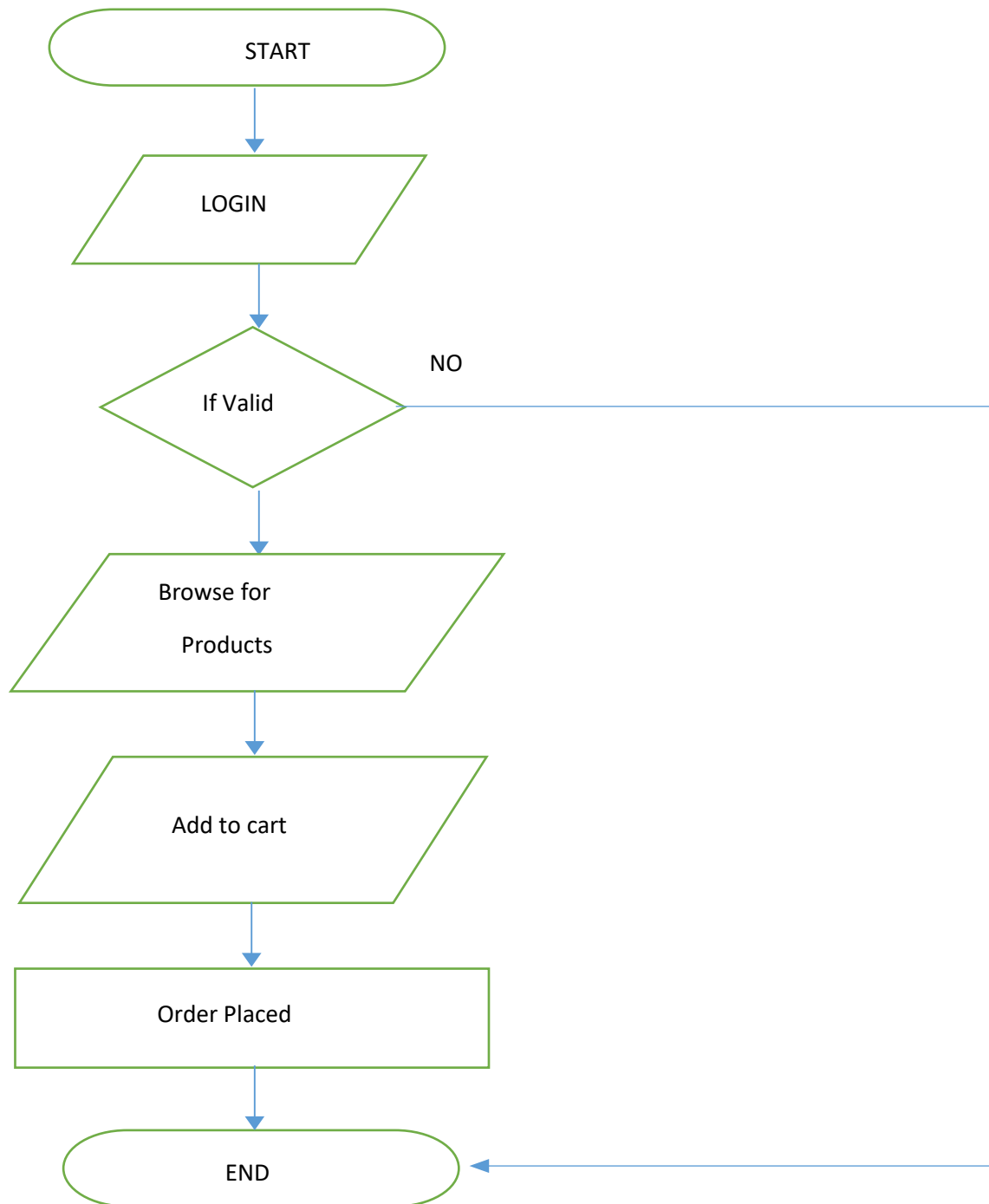


fig. Customer Flow Chart

Interface Design

Interface design plays a critical role in the usability and success of any web application. In the Online E-Commerce System, the interface has been designed to be clean, intuitive, and responsive, ensuring ease of access and navigation for all types of users — including buyers, sellers, and the admin.

Design Principles Followed:

- **Simplicity:**
The layout is kept minimal to avoid confusion. Each page focuses on a single purpose (e.g., login page, product detail page, cart).
- **Consistency:**
Colour schemes, button styles, fonts, and layouts remain consistent throughout the application.
- **Responsiveness:**
The system is designed using responsive HTML and CSS, allowing it to work seamlessly on mobile, tablet, and desktop screens.
- **Accessibility:**
Proper font sizes, contrasting colours, and alt texts for images are used to enhance accessibility for all users.

Major Interfaces:

1. **Home Page:**
 - Displays featured and latest products.
 - Includes a search bar and navigation menu.
 - Promotional banners (optional) can be added.
2. **User Registration/Login Pages:**
 - Separate login/registration for buyers and sellers.
 - Secure login with input validation and error prompts.
3. **Admin Dashboard:**
 - Complete control panel with sections for:
 - User management
 - Product & category management
 - Order tracking
 - Discount and offer management
 - Displays statistics and recent activities.
4. **Product Listing Pages:**
 - Category-wise and keyword-filtered listing.
 - Product cards show image, name, price, and "Add to Cart" button.
5. **Product Detail Page:**
 - Larger image, full description, price, stock availability, and add-to-cart option.
6. **Shopping Cart & Checkout:**
 - Simple cart interface with quantity update, remove item, and total calculation.
 - Checkout page with form for shipping info and order confirmation.
7. **User Profile Panel:**
 - Displays order history, personal info, and password change options.

DESIGN PHASE

DETAILED LIFE CYCLE OF PROJECT

- Event table

Sr.No.	Event	Trigger	Source	Activity	Response	Destination
1	Admin Login	Login	Admin	Checks for valid username and password	Opens the Admin page if successful	Admin
2	Admin Logout	Logout	Admin	Logout	The login page is opened.	Admin
3	Add customer	Entering customer details	Registration Page	Adding new Customer to the record.	New customer gets added	Admin
4	Edit Admin	Entering Admin details	Admin	Editing/ adding admin	Admin details get edited	Admin
5	Delete customer	Delete customer	Admin	Deleting customer	Deleted customer	Admin
6	Update/Del etc/Add product	Entering Product details	Admin	Adding product	Product gets added	Admin
7	Customer login	Login	Customer	Checks for valid email id and password	Opens the cart process if successful	Customer
8	Customer logout	Logout	Customer	Logs out of the session	The main homepage is opened	Customer

Chapter 5: Implementation

TOOLS AND TECHNOLOGY:

The tools that we are using for this software are:

- **XAMPP:**

XAMPP is a free and open-source cross-platform webserver solution stack package developed by Apache Friends, consisting mainly of the Apache HTTP Server, MariaDB database, and interpreters for scripts written in the PHP and Perl programming languages, for now we require Apache and MySQL only.

- **Apache Server:**

- The Apache HTTP Server Project is an effort to develop and maintain an opensource HTTP server for modern operating systems including UNIX and Windows. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.
- The Apache HTTP Server ("httpd") was launched in 1995 and it has been the most popular web server on the Internet since April 1996. It has celebrated its 20th birthday as a project in February 2015.

- **MySQL:**

- MySQL is an open-source relational database management system (RDBMS). Its name is a combination of "My", the name of co-founder Michael Widenius's daughter, and "SQL", the abbreviation for Structured Query Language. The MySQL development project has made its source code available under the terms of the GNU General Public License, as well as under a variety of proprietary agreements. MySQL was owned and sponsored by a single for-profit firm, the Swedish company MySQL AB, now owned by Oracle Corporation

Technologies that we have used in our project are as follows:

- MySQL
- PHP
- HTML
- CS
- Java Script

- **Back End**

- **SQL**

- MySQL is a database system used on the web
 - MySQL is a database system that runs on a server
 - MySQL is ideal for both small and large applications
 - MySQL is very fast, reliable, and easy to use
 - MySQL uses standard SQL.

- **PHP**

- PHP is an acronym for "PHP: Hypertext Preprocessor"
 - PHP is a server scripting language, and a powerful tool for making dynamic and interactive Web pages.
 - PHP is a widely-used, free, and efficient alternative to competitors such as Microsoft's ASP.
 - PHP is free to download and use

- **Front End**

- **HTML:**

- HyperText Markup Language, commonly referred to as HTML, is the standard markup language used to create web pages. It is written in the form of HTML elements consisting of *tags* enclosed in angle brackets.
 - For creating any website or web application we have the basic thing required is HTML.

- **CSS:**

- CSS classes using which one can develop more interactive websites and web application.
 - We have used CSS classes to make our website look more user attractive, better component stabilization and using this our web application can be viewed in any mobiles or tab.

Database Integration

components > connect.php > ...

```
1  <?php
2  // filepath: c:\xampp\htdocs\projectdone\components\connect.php
3  // No UI elements here, but for consistency, you can wrap DB errors in a Bootstrap alert if needed.
4
5  $db_name = 'mysql:host=localhost;dbname=shop_db';
6  $user_name = 'root';
7  $user_password = '';
8
9  try {
10     $conn = new PDO(dsn: $db_name, username: $user_name, password: $user_password);
11     $conn->setAttribute(attribute: PDO::ATTR_ERRMODE, value: PDO::ERRMODE_EXCEPTION);
12 } catch(PDOException $e) {
13     // Simple Bootstrap 5 alert for DB connection error
14     echo '
15     <div class="alert alert-danger m-3" role="alert" style="border-radius:2rem;">
16         <strong>Database Connection Failed:</strong> ' . htmlspecialchars(string: $e->getMessage()) . '
17     </div>
18     ';
19     exit;
20 }
21 ?>
```

The screenshot shows the phpMyAdmin interface with the 'shop_db' database selected. The left sidebar shows the database structure, including tables like 'admins', 'cart', 'messages', 'orders', 'products', 'users', and 'wishlist'. The main area displays a table structure view for 'shop_db' with columns: Table, Action, Rows, Type, Collation, Size, and Overhead. The table 'admins' is highlighted.

Table	Action	Rows	Type	Collation	Size	Overhead
admins	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	16.0 K1B	-
cart	Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	16.0 K1B	-
messages	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	16.0 K1B	-
orders	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	16.0 K1B	-
products	Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	16.0 K1B	-
users	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	16.0 K1B	-
wishlist	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 K1B	-
7 tables	Sum	18	InnoDB	utf8mb4_general_ci	112.0 K1B	0 B

APIs / Modules Used

To make the system smarter and more interactive, several core modules were developed along with one powerful third-party API — Google Gemini API, which powers an AI-based chatbot assistant for users.

1. Gemini API (Chatbot Integration):

- **Purpose:**

Gemini API is used to provide intelligent chatbot support for users within the website. This chatbot assists users with product queries, navigation help, order issues, and FAQs.

- **Integration:**

The chatbot is embedded using JavaScript + PHP, and communicates with Gemini's API endpoints via HTTP requests.

- **Features Provided:**

- o Instant product-related answers
- o Helps in tracking orders or returns
- o User support 24/7 without human admin
- o Suggests products based on user's input
- o Can answer FAQs about registration, login, etc.

- **Advantages:**

- o Improves user engagement
- o Reduces admin workload
- o Adds a modern, AI-powered touch to the e-commerce platform

2. System Modules (Developed in PHP):

The project is modular in nature, consisting of various independent but interconnected components:

Module Name	Description
User Module	Handles user registration, login, and profile management
Admin Module	Allows admin to manage users, products, categories, and orders
Product Module	Enables dynamic product listing, filtering, and detail viewing
Cart & Checkout Module	Manages shopping cart items, calculates totals, and processes checkouts
Order Management	Stores order data, tracks status, and allows user/admin to monitor progress
Search Module	Supports keyword + filter-based product searching
Discount Module	Admin can apply discounts to specific products or categories
Chatbot Module	Integrated with Gemini API to provide instant support

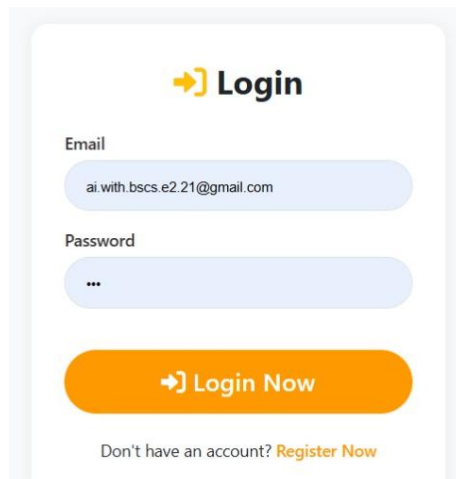
Tools & Technologies for Modules:

- Backend: PHP 7+, MySQL
- Frontend: HTML5, CSS3, Bootstrap
- AJAX: For seamless background requests (e.g., add to cart without reloading)
- JavaScript: For UI enhancements and API communication
- Gemini API: For AI-powered user interaction

Screenshots of the Working System

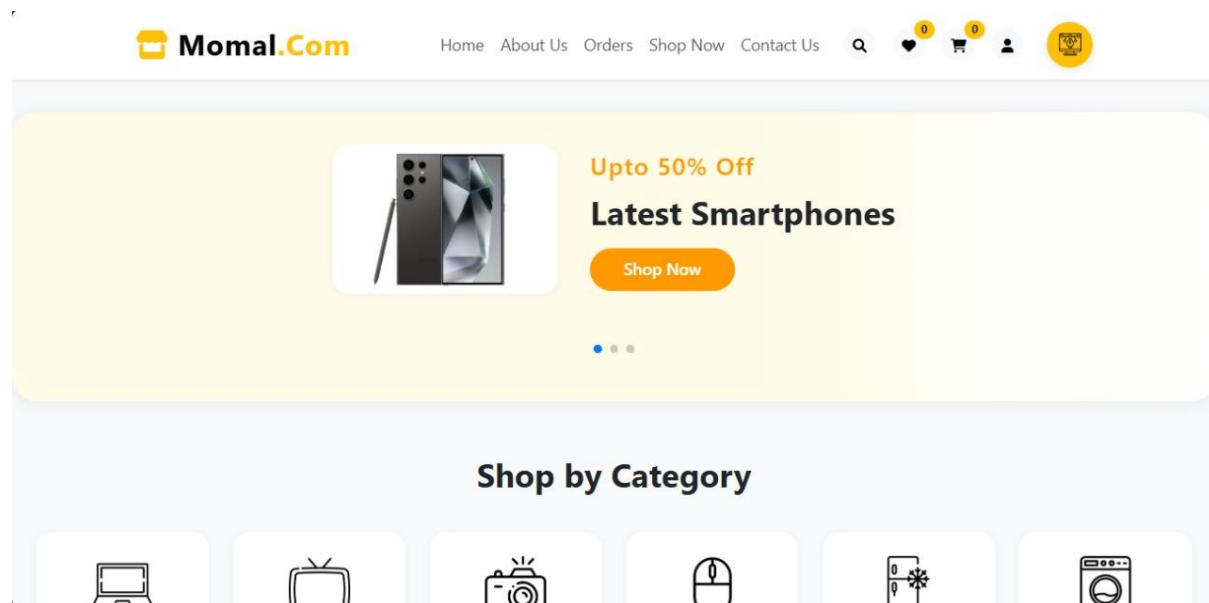
Customers Side

Login Screen



The login screen features a white card with a light gray border. At the top, there is a yellow icon of a right-pointing arrow inside a square bracket, followed by the word "Login" in bold black text. Below this, there are two input fields: "Email" with the placeholder text "ai.with.bscs.e2.21@gmail.com" and "Password" with three dots as a placeholder. Both fields have a light blue background. Below the input fields is a large orange button with a white right-pointing arrow icon and the text "Login Now". At the bottom of the card, there is a link that says "Don't have an account? Register Now" in a smaller, lighter orange font.

Dashboard



Products

 **Latest Products**



nothing smartphone
3
RS.2000000/-
1
Add to Cart



i phone 13
RS.1200000/-
1
Add to Cart



laptop i5 4th gen
RS.30000/-
1
Add to Cart



DSLR Camera
RS.90000/-
1
Add to Cart

About Section

 Home About Us Orders Shop Now Contact Us     



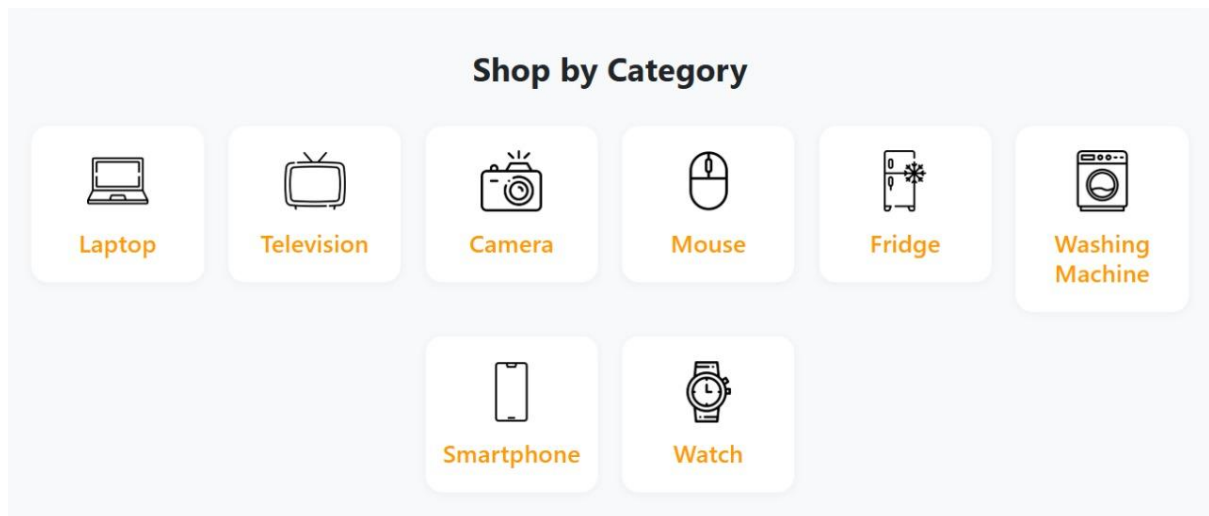
Developer's Message

Hey There! I'm **Wajeeha Batool**, a BSCS student at University of Sahiwal (Batch: 2021-25). I love designing web apps and exploring new things. Learning is my hobby!

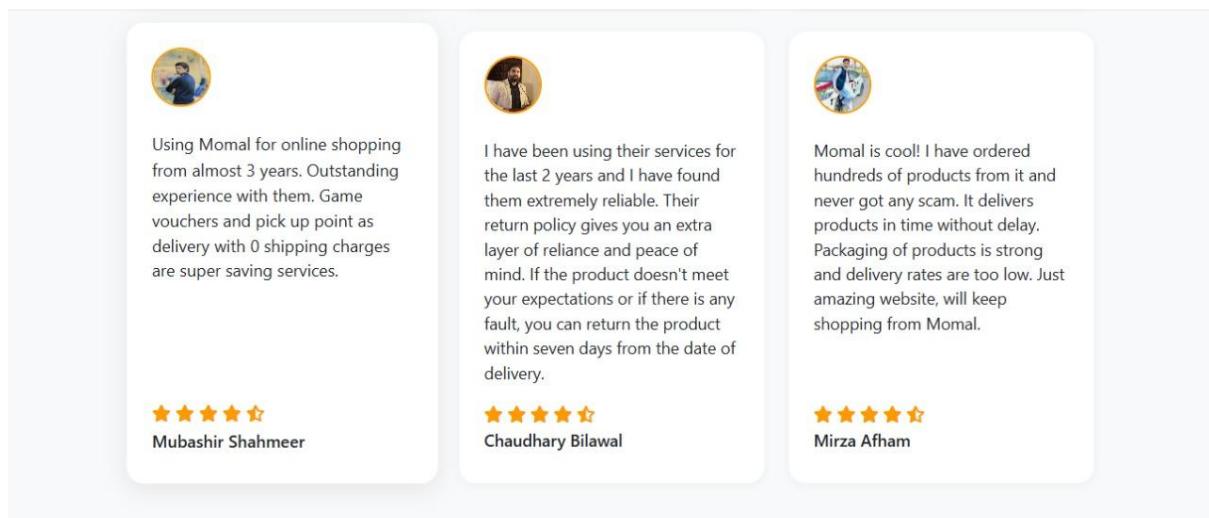
I would like to thank **Umer Daraz** Sir for guiding me and making me able to develop projects like this.

[Contact Us](#)

Category



Customers Feedback



Orders Section

 **Placed Orders**



Please login to see your orders

Message

 **Get in Touch**

Your Name

Enter your name

Your Email

Enter your email

Phone Number

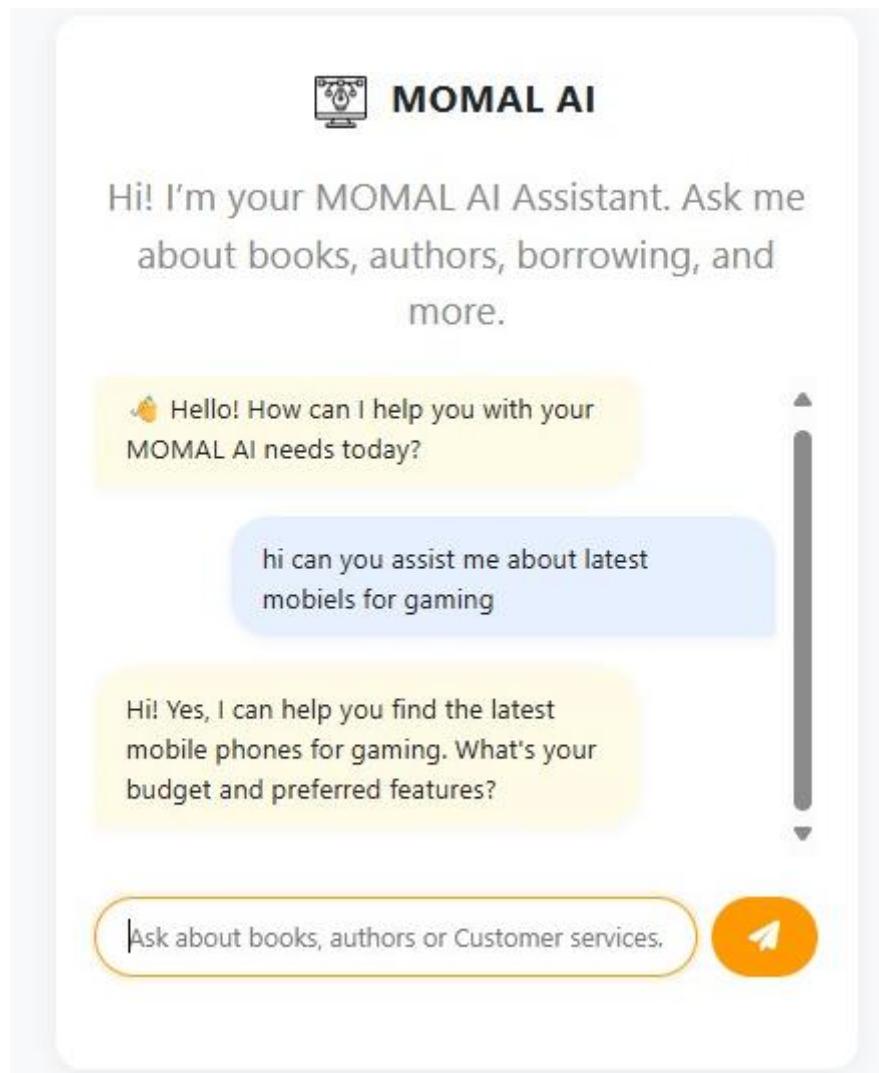
Enter your number

Message

Enter your message

 **Send Message**

AI Chatbot



Contact us

**Address**
Sahiwal, Punjab, Pakistan

**Email**
info@momal.com

**Phone**
+92 300 0000000

Quick Links

- > Home
- > About
- > Shop
- > Contact

Extra Links

- > Login
- > Register
- > Cart
- > Orders

Contact Us

- +92 316 6336455
- +92 312 0000000
- Wajeeha001@gmail.com
- Sahiwal, Punjab

Follow Us

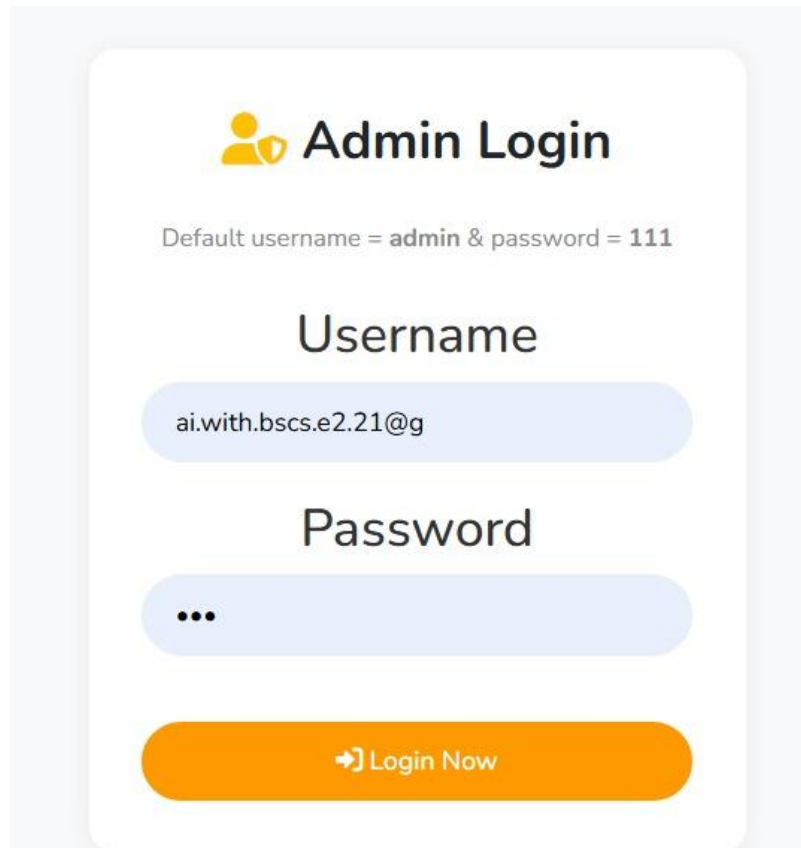
- Facebook
- GitHub
- Instagram
- LinkedIn

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Screenshots of the Working System

Admin Side

Login Screen



The image shows a login screen for an admin user. It features a white card with rounded corners on a light gray background. At the top, there is a yellow icon of a person with a shield, followed by the text "Admin Login". Below this, a message states "Default username = admin & password = 111". The form has two input fields: "Username" with the value "ai.with.bscs.e2.21@g" and "Password" with masked characters "...". At the bottom, there is an orange button with a right arrow icon and the text "Login Now".

Admin Login

Default username = admin & password = 111

Username

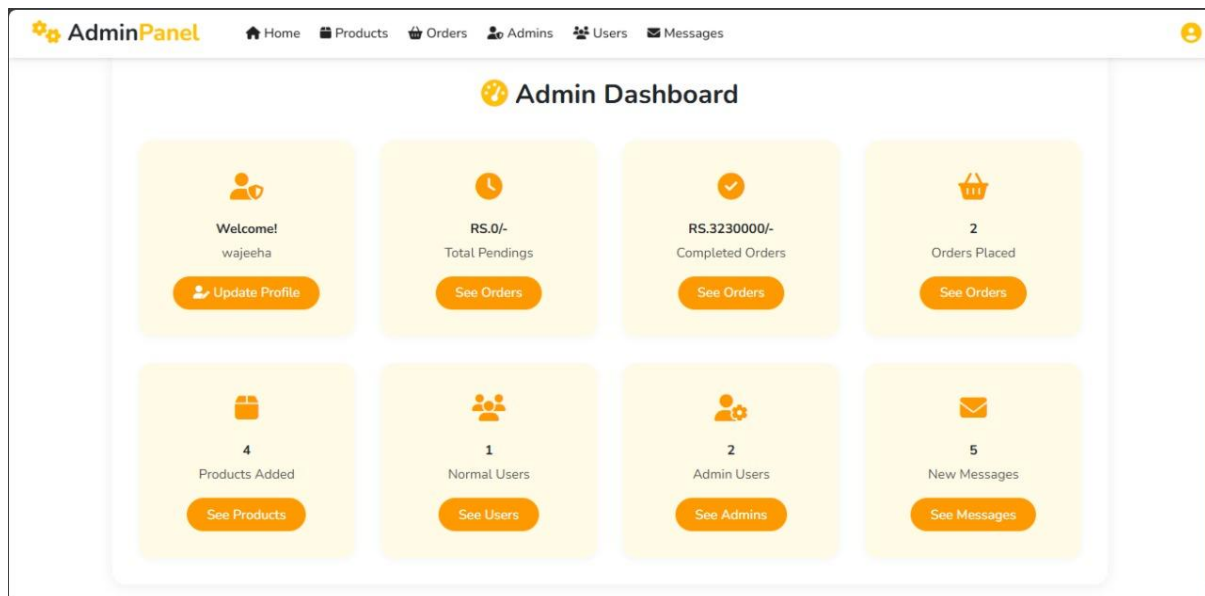
ai.with.bscs.e2.21@g

Password

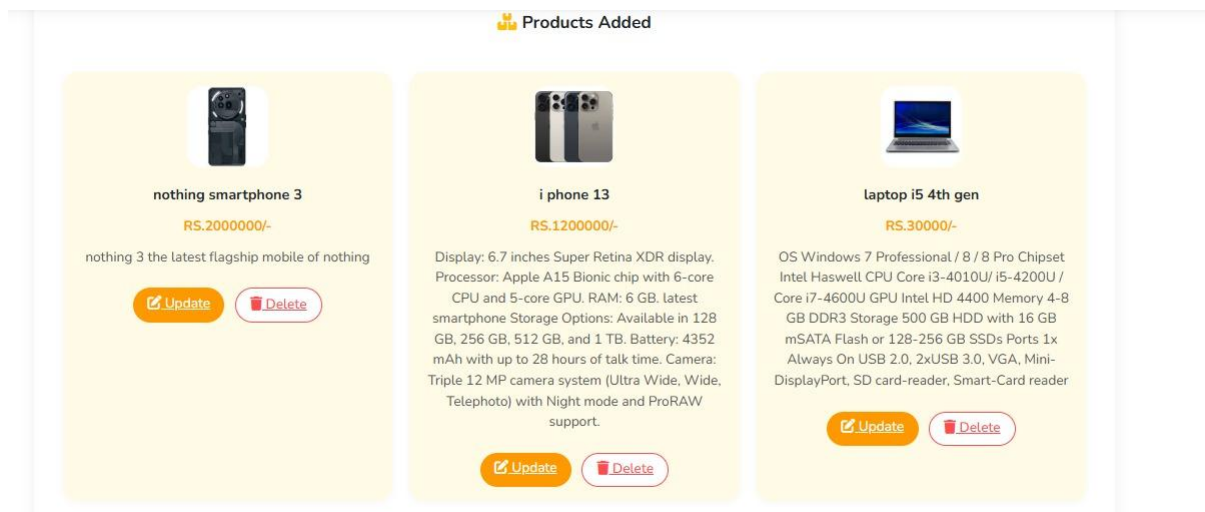
...

➔ Login Now

Dashboard



Products List



Add Product

 Add Product

Product Name *

Enter product name

Product Price *

Enter product price

Image 01 *

Choose File No file chosen

Image 02 *

Choose File No file chosen

Image 03 *

Choose File No file chosen

Product Description *

Enter product details

+ Add Product

Orders Details

 Placed Orders

 Placed On: 2025-06-26

 Name: Kahaf Sameer

 Number: 0300840257

 Address: flat no. ShamsPura CCW, shampura CCW, CCW, punjab, Pakistan - 57000

 Total Products: nothing smartphone 3 (2000000 x 1) - i phone 13 (1200000 x 1) -

 Total Price: RS.3200000/-

 Payment Method: cash on delivery

completed

Update

Delete

 Placed On: 2025-06-27

 Name: sameer

 Number: 987798689

 Address: flat no. ShamsPura CCW, shampura CCW, CCW, punjab, Pakistan - 62

 Total Products: laptop i5 4th gen (30000 x 1) -

 Total Price: RS.300000/-

 Payment Method: credit card

completed

Update

Delete

E-Commerce Website

65 | Page

Admins List

 **Admin Accounts**

 Add New Admin

Admin ID: 1

 admin

 [Delete](#)

Admin ID: 2

 wajeetha

 [Delete](#)  [Update](#)

Users Account

 **User Accounts**

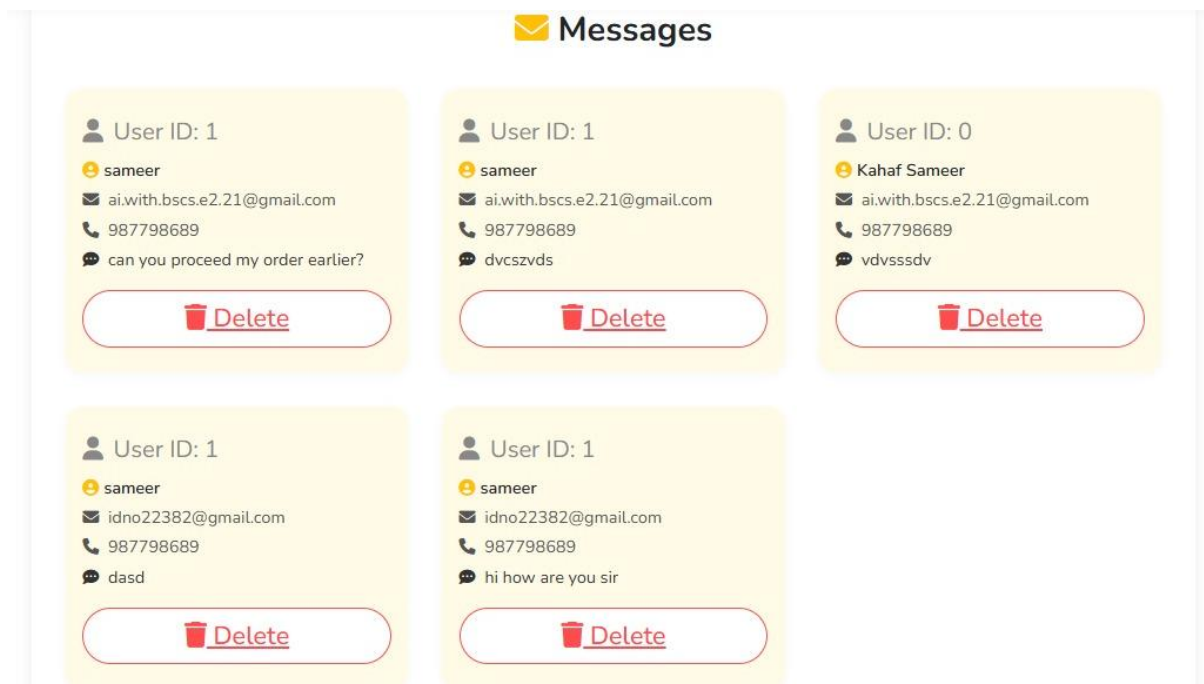
 User ID: 1

 Username: wajeetha

 Email: ai.with.bscs.e2.21@gmail.com

 [Delete](#)

Message Inbox



Check Profile



Update Admin Profile

 **Update Profile**

Username

wajeeha

Old Password

...

New Password

Enter new password

Confirm New Password

Confirm new password

 Update Now

Chapter 6: Testing & Evaluation

Testing Strategy (Unit, Integration, System)

UNIT TESTING

Introduction

In computer programming, unit testing is a software testing method by which individual units of source code, sets of one or more computer program modules together with associated control data, usage procedures, and operating procedures, are tested to determine whether they are fit for use. Intuitively, one can view a unit as the smallest testable part of an application. In procedural programming, a unit could be an entire module, but it is more commonly an individual function or procedure. In object-oriented programming, a unit is often an entire interface, such as a class, but could be an individual method. Unit tests are short code fragments created by programmers or occasionally by white box testers during the development process. It forms the basis for component testing. Ideally, each test case is independent from the others. Substitutes such as method stubs, mock objects, fakes, and test harnesses can be used to assist testing a module in isolation. Unit tests are typically written and run by software developers to ensure that code meets its design and behaves as intended.

Benefits

The goal of unit testing is to isolate each part of the program and show that the individual parts are correct. A unit test provides a strict, written contract that the piece of code must satisfy. As a result, it affords several benefits.

1) Find problems early: Unit testing finds problems early in the development cycle. In test-driven development (TDD), which is frequently used in both extreme programming and scrum, unit tests are created before the code itself is written. When the tests pass, that code is considered complete. The same unit tests are run against that function frequently as the larger code base is developed either as the code is changed or via an automated process with the build. If the unit tests fail, it is considered to be a bug either in the changed code or the tests themselves. The unit tests then allow the location of the fault or failure to be easily traced. Since the unit tests alert the development team of the problem before handing the code off to testers or clients, it is still early in the development process.

2) Facilitates Change: Unit testing allows the programmer to refactor code or upgrade system libraries at a later date, and make sure the module still works correctly (e.g., in regression testing). The procedure is to write test cases for all functions and methods so that whenever a change causes a fault, it can be quickly identified. Unit tests detect changes which may break a design contract.

3) Simplifies Integration: Unit testing may reduce uncertainty in the units themselves and can be used in a bottom-up testing style approach. By testing the parts of a program first and then testing the sum of its parts, integration testing becomes much easier.

4) Documentation: Unit testing provides a sort of living documentation of the system. Developers looking to learn what functionality is provided by a unit, and how to use it, can look at

the unit tests to gain a basic understanding of the unit's interface (API). Unit test cases embody characteristics that are critical to the success of the unit. These characteristics can indicate appropriate/inappropriate use of a unit as well as negative behaviors that are to be trapped by the unit.

INTEGRATION TESTING

Integration testing (sometimes called integration and testing, abbreviated I&T) is the phase in software testing in which individual software modules are combined and tested as a group. It occurs after unit testing and before validation testing. Integration testing takes as its input modules that have been unit tested, groups them in larger aggregates, applies tests defined in an integration test plan to those aggregates, and delivers as its output the integrated system ready for system testing.

PURPOSE

The purpose of integration testing is to verify functional, performance, and reliability requirements placed on major design items. These "design items", i.e., assemblages (or groups of units), are exercised through their interfaces using black-box testing, success and error cases being simulated via appropriate parameter and data inputs. Simulated usage of shared data areas and inter-process communication is tested and individual subsystems are exercised through their input interface. Test cases are constructed to test whether all the components within assemblages interact correctly, for example across procedure calls or process activations, and this is done after testing individual modules, i.e., unit testing. The overall idea is a "building block" approach, in which verified assemblages are added to a verified base which is then used to support the integration testing of further assemblages. Software integration testing is performed according to the software development life cycle (SDLC) after module and functional tests. The cross dependencies for software integration testing are: schedule for integration testing, strategy and selection of the tools used for integration, define the cyclomatic complexity of the software and software architecture, reusability of modules and life-cycle and versioning management. Some different types of integration testing are big-bang, top-down, and bottom-up, mixed (sandwich) and risky-hardest. Other Integration Patterns [2] are: collaboration integration, backbone integration, layer integration, client-server integration, distributed services integration and high-frequency integration.

Big Bang

In the big-bang approach, most of the developed modules are coupled together to form a complete software system or major part of the system and then used for integration testing. This method is very effective for saving time in the integration testing process. However, if the test cases and their results are not recorded properly, the entire integration process will be more complicated and may prevent the testing team from achieving the goal of integration testing. A type of big-bang integration testing is called "usage model testing" which can be used in both software and hardware integration testing. The basis behind this type of integration testing is to run user-like workloads in integrated user-like environments. In doing the testing in this manner, the environment is proofed, while the individual components are proofed indirectly through their use. Usage Model testing takes an optimistic approach to testing, because it expects to have few

problems with the individual components. The strategy relies heavily on the component developers to do the isolated unit testing for their product. The goal of the strategy is to avoid redoing the testing done by the developers, and instead flesh-out problems caused by the interaction of the components in the environment.

Top-down And Bottom-up

Bottom-up testing is an approach to integrated testing where the lowest level components are tested first, then used to facilitate the testing of higher-level components. The process is repeated until the component at the top of the hierarchy is tested. All the bottom or low-level modules, procedures or functions are integrated and then tested. After the integration testing of lower-level integrated modules, the next level of modules will be formed and can be used for integration testing. This approach is helpful only when all or most of the modules of the same development level are ready. This method also helps to determine the levels of software developed and makes it easier to report testing progress in the form of a percentage. Top-down testing is an approach to integrated testing where the top integrated modules are tested and the branch of the module is tested step by step until the end of the related module. Sandwich testing is an approach to combine top-down testing with bottom-up testing.

Test Cases

TEST CASES

Black Box Testing Technique: Boundary Value Analysis (BVA)

Test Cases:1

Identifier: T1

Purpose: Test functionality of login form.

Pre-requisite: Appropriate database with sample record should be ready.

Test Input/data:

Test Data/Action	Condition Description	Excepted Result	Actual Result	Remark
T01.Email Id& password Validation	1.Either Email Id or Password both wrong.	It should display Error Message “Password or email is incorrect, please try again!”.	It displays Error Message “Password or email is incorrect, please try again!”.	Pass
	2.Enter null value for Email Id.	It should display Error Message “Please fill out this field”.	It displays Error Message “Please fill out this field”.	Pass
	3.Enter null value for Password.	It should display Error Message “Please fill out this field”.	It displays Error Message “Please fill out this field”.	Pass
T02.Submit	Click on Submit.	It should validate the fields and navigate to proper page.	It validates the fields and navigate to proper page.	Pass
T04.Don’t have account then please Register yourself.	Click on the link.	Should Navigate to register page.	It navigates to register page.	Pass

Test Cases:2

Identifier: 2

Purpose: Test functionality of Registration form.

Pre-requisite: Appropriate database with sample record should be ready.

Test Input/data:

Test Data/Action	Condition Description	Excepted Result	Actual Result	Remark
T01.Blank form validation		Form should not be submitted & all the validation errors are displayed.	Form is not submitted & all the validation errors are displayed.	Pass
T02. Empty (Null value) fields validation	1.Enter null value for first name.	It should display error message “Please fill out this field”.	It displays error message “Please fill out this field”.	Pass
	2.Enter null value for last name.	It should display error message “Please fill out this field”.	It displays error message “Please fill out this field”.	Pass
	3.Enter null value for password.	It should display error message “Please fill out this field”.	It displays error message “Please fill out this field”.	Pass
	4.Enter null value for contact no.	It should display error message “Please fill out this field”.	It displays error message “Please fill out this field”.	Pass
	5.Enter null value for Email Id.	It should display error message “Please fill out this field”.	It displays error message “Please fill out this field”.	Pass

	no	Contact no field should have only 10 or 13 digits.	It should display error message “Please match the requested format: Minimum 10 digits”.	It displays error message “Please match the requested format: Minimum 10 digits”.	Pass
T05.Email validation	Id	1.Invalid Email Id format	It should display error message “Please Enter an email address”.	It displays error message "Please enter an email address".	Pass
T06.Submit		Click on Submit.	It should validate the fields and navigate to proper page.	It validates the fields and navigate to proper page.	Pass
		6.Enter null value for address.	It should display error message “Please fill out this field”.	It displays error message Please fill out this field”.	Pass

Test Cases:3

Identifier: 3

Purpose: Check whether different hyperlinks are working properly or not.

Test Data/Action	Condition Description	Excepted Result	Actual Result	Remark
T01.click on Home page hyperlink menu	-	Should display home page	It displays Home page	Pass
T02.click on All Products us page hyperlink menu	-	Should display all products page	It displays all products page	Pass
T03.click on My Account page hyperlink menu	-	Should display My Account page	It displays My Account page	Pass
T04.click on Brands page hyperlink menus	-	Should display all Brands	It displays all Brands	Pass
T05.click on Contact us page hyperlink menu	-	Should display Contact us page	It displays Contact us page	Pass
T06.click on Cart page hyperlink menu	-	Should display Cart page	It displays Cart page	Pass

Test Cases:4

Identifier: T04 Purpose:

Check whether the product purchase transaction by the Customer/shopkeeper takes place smoothly.

Pre-requisite: Appropriate database with sample record should be ready.

Test Data/Action	Condition Description	Excepted Result	Actual Result	Remark
T01.Add to cart	Click add to cart button.	It should open cart page & add product to cart.	It opens cart page & add product to cart.	Pass
T02.Products with total amount, cost.	Click on checkout button.	It should go to the PayPal checkout page.	It goes to the PayPal checkout page.	Pass

Results & Observations

After implementing and testing the Online E-Commerce System, several results and behavioural observations were recorded. These results validate the system's functional correctness, ease of use, and effectiveness in a simulated real-world environment.

Functional Results:

- **User Registration:**
Both buyers and sellers can register successfully. Seller accounts require admin approval before becoming active.
- **Product Management:**
Admin can easily add, update, and delete product listings. Products appear dynamically on the homepage and category pages.
- **Search and Filter:**
The system successfully filters products by category, keyword, and price range — improving product discovery and user experience.
- **Order Placement:**
Registered buyers can place orders, and a simple order tracking system allows monitoring of purchase history.
- **Discounts and Offers:**
Admin-added discounts are displayed automatically on product pages, and prices are adjusted accordingly.

Observations:

- The system loads quickly on most browsers and works on both desktop and mobile devices.
- All major functionalities worked without errors in the test environment.
- Users responded positively to the simplicity and clarity of the UI.
- Proper validations prevented most types of invalid inputs.
- Minor layout issues were observed on older mobile browsers — which can be fixed in future updates.

Performance Evaluation

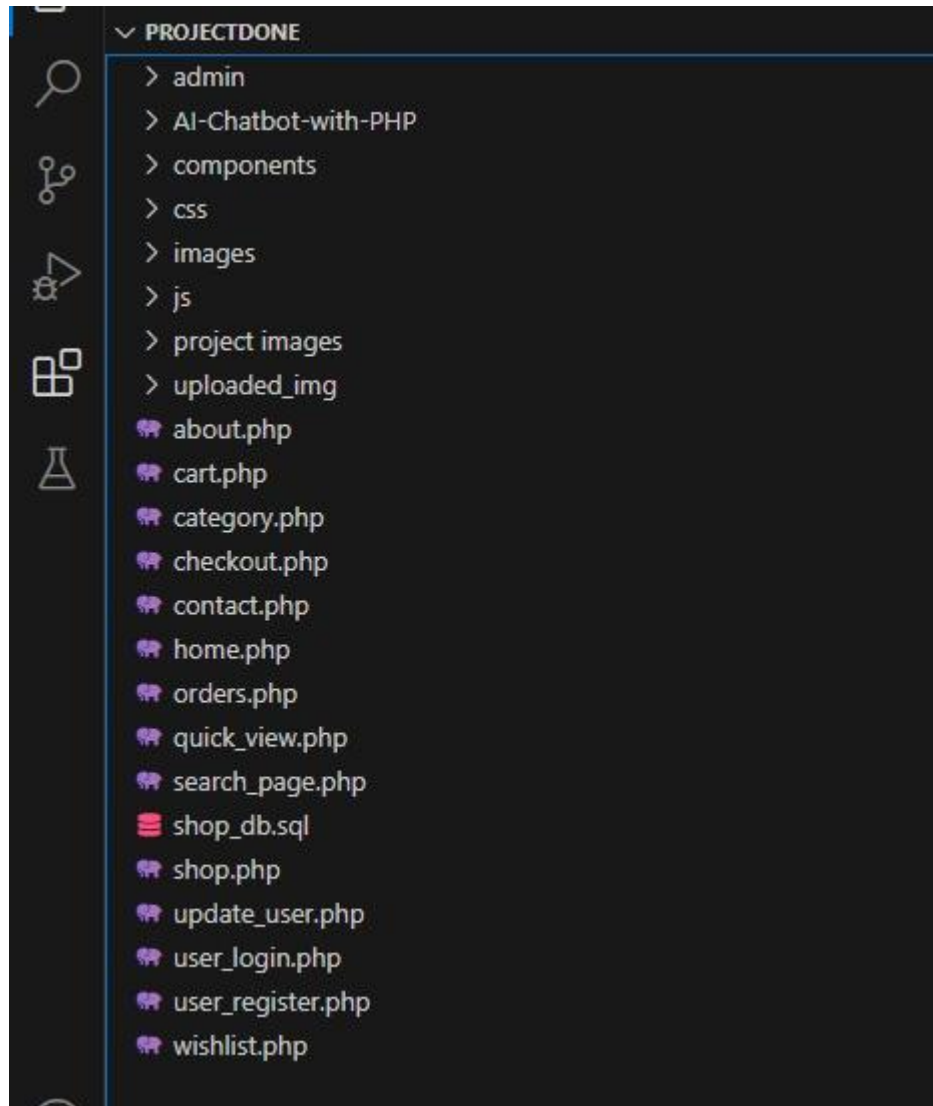
The performance of the system was evaluated based on factors like speed, responsiveness, data handling, and resource usage during various operations. Testing was conducted in a local server environment using XAMPP (PHP and MySQL).

Evaluation Parameters:

Metric	Observation
Page Load Time	0.5 – 1.5 seconds (average)
Search Query Response	< 1 second for keyword & filtered search
Database Transactions	Stable under 500+ concurrent reads/writes
Form Validation	Fast response with accurate error handling
Mobile Responsiveness	100% responsive on latest devices and browsers
Resource Usage	Low CPU and RAM usage on shared hosting
Scalability	Designed to handle moderate traffic easily

Code Integrated

Project Structure



Admin Header

```
... home.php shop.php quick_view.php admin_header.php orders.php cart.php contact.php
header.php > nav.navbar.navbar-expand-lg.navbar-light.bg-white.sticky-top.shadow-sm > div.container-fluid > div#adminNavbar.collapse.navbar-collapse > div.d-flex
15 class="navbar navbar-expand-lg navbar-light bg-white sticky-top shadow-sm" style="z-index: 1030;"
16 class="container-fluid">
19 a>
20 utton class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#adminNavbar" aria-controls="adminNavb
21 <span class="navbar-toggler-icon"></span>
22 button>
23 iv class="collapse navbar-collapse" id="adminNavbar">
24 <ul class="navbar-nav me-auto mb-2 mb-lg-0">
25 <li class="nav-item"><a class="nav-link" href="../admin/dashboard.php">
26 <i class="fas fa-home"></i> Home</a></li>
27 <li class="nav-item"><a class="nav-link" href="../admin/products.php">
28 <i class="fas fa-box"></i> Products</a></li>
29 <li class="nav-item"><a class="nav-link" href="../admin/placed_orders.php">
30 <i class="fas fa-shopping-basket"></i> Orders</a></li>
31 <li class="nav-item"><a class="nav-link" href="../admin/admin_accounts.php">
32 <i class="fas fa-user-shield"></i> Admins</a></li>
33 <li class="nav-item"><a class="nav-link" href="../admin/users_accounts.php">
34 <i class="fas fa-users"></i> Users</a></li>
35 <li class="nav-item"><a class="nav-link" href="../admin/messages.php">
36 <i class="fas fa-envelope"></i> Messages</a></li>
37 </ul>
38 <div class="d-flex align-items-center">
39 <?php
40 $select_profile = $conn->prepare(query: "SELECT * FROM `admins` WHERE id = ?");
41 $select_profile->execute(params: [$admin_id]);
42 $fetch_profile = $select_profile->fetch(mode: PDO::FETCH_ASSOC);
43 ?>
44 <div class="dropdown">
45 <a class="btn btn-light rounded-pill px-3 fw-semibold dropdown-toggle d-flex align-items-center" href="#" role="button"
46 <i class="fas fa-user-circle me-2 text-warning"></i>
47 <?= htmlspecialchars(string: $fetch_profile['name']); ?>
48 </a>
```

Footer

```
home.php footer.php orders.php cart.php contact.php
components > footer.php > footer.footer.bg-white.border-top.mt-5.pt-4 > style > .footer-link
1 <footer class="footer bg-white border-top mt-5 pt-4">
2 <div class="container">
3 <div class="row g-4 text-center text-md-start">
4 <div class="col-12 col-sm-6 col-md-3">
5 <h5 class="fw-bold mb-3 text-warning">Quick Links</h5>
6 <ul class="list-unstyled">
7 <li><a href="home.php" class="footer-link"><i class="fas fa-angle-right me-2"></i>Home</a></li>
8 <li><a href="about.php" class="footer-link"><i class="fas fa-angle-right me-2"></i>About</a></li>
9 <li><a href="shop.php" class="footer-link"><i class="fas fa-angle-right me-2"></i>Shop</a></li>
10 <li><a href="contact.php" class="footer-link"><i class="fas fa-angle-right me-2"></i>Contact</a></li>
11 </ul>
12 </div>
13 <div class="col-12 col-sm-6 col-md-3">
14 <h5 class="fw-bold mb-3 text-warning">Extra Links</h5>
15 <ul class="list-unstyled">
16 <li><a href="user_login.php" class="footer-link"><i class="fas fa-angle-right me-2"></i>Login</a></li>
17 <li><a href="user_register.php" class="footer-link"><i class="fas fa-angle-right me-2"></i>Register</a></li>
18 <li><a href="cart.php" class="footer-link"><i class="fas fa-angle-right me-2"></i>Cart</a></li>
19 <li><a href="orders.php" class="footer-link"><i class="fas fa-angle-right me-2"></i>Orders</a></li>
20 </ul>
21 </div>
22 <div class="col-12 col-sm-6 col-md-3">
23 <h5 class="fw-bold mb-3 text-warning">Contact Us</h5>
24 <ul class="list-unstyled">
25 <li><a href="tel:03166336455" class="footer-link"><i class="fas fa-phone me-2"></i>+92 316 6336455</a></li>
26 <li><a href="tel:03120000000" class="footer-link"><i class="fas fa-phone me-2"></i>+92 312 0000000</a></li>
27 <li><a href="mailto:ahmad001@gmail.com" class="footer-link"><i class="fas fa-envelope me-2"></i>ahmad001@gmail.com</a></li>
28 <li><a href="https://www.google.com/myplace" class="footer-link" target="_blank"><i class="fas fa-map-marker-alt me-2"></i>Sahiwal
29 </ul>
30 </div>
31 <div class="col-12 col-sm-6 col-md-3">
32 <h5 class="fw-bold mb-3 text-warning">Follow Us</h5>
```

Cart

```
home.php about.php orders.php cart.php x contact.php
cart.php > html > body > div.container.my-5 > div.cart-section > div.row.g-4
40 <html lang="en">
147 <body>
152 <div class="container my-5">
153 <div class="cart-section">
155 <div class="row g-4">
156 <div class="col-12 col-md-6 col-lg-4">
164 <div class="card cart-item-card h-100">
165 <input type="hidden" name="cart_id" value="<?=$fetch_cart['id']; ?>">
167 <div class="card-body d-flex flex-column align-items-center">
168 <a href="quick_view.php?pid=<?=$fetch_cart['pid']; ?>" class="mb-2">
169 " class="cart-item-img shadow">
170 </a>
171 <div class="card-item-name mt-2 mb-1 text-center"><?=$fetch_cart['name']; ?></div>
172 <div class="card-item-price mb-2">RS.<?=$fetch_cart['price']; ?></div>
173 <div class="card-item-actions d-flex align-items-center gap-2 mb-2">
174 <input type="number" name="qty" class="form-control text-center" style="width:70px;" min="1" max="99" onkeypress="if(this.va
175 <button type="submit" class="btn btn-outline-warning" name="update_qty" title="Update Quantity"><i class="fas fa-edit"></i>
176 </div>
177 <div class="mb-2 text-muted">Sub Total: <span class="fw-bold">RS.<?=$sub_total; ?></span></div>
178 <button type="submit" onclick="return confirm('Delete this from cart?');" class="btn btn-danger w-100" name="delete"><i class="
179 </div>
180 </form>
181 </div>
182 </div>
183 <?php
184 <div class="empty-cart w-100"><i class="fas fa-shopping-basket fa-2x mb-3"></i><br>Your cart is empty</div>;
185 }else{
186 echo '
187 }
188 </div>
189 </div>
```

Home file

```
home.php x
home.php > html > body > div.container.products-section > div.row.g-4 > div.col-12.col-sm-6.col-md-4.col-lg-4
18 <html lang="en">
244 <body>
288 <!-- Category Section -->
289 <div class="container category-section">
290 <div class="category-title">Shop by Category</div>
291 <div class="row g-4 justify-content-center">
292 <div class="col-6 col-sm-4 col-md-3 col-lg-2">
293 <a href="category.php?category=laptop" class="category-card d-block text-decoration-none">
294 
295 <h3>Laptop</h3>
296 </a>
297 </div>
298 <div class="col-6 col-sm-4 col-md-3 col-lg-2">
299 <a href="category.php?category=tv" class="category-card d-block text-decoration-none">
300 
301 <h3>Television</h3>
302 </a>
303 </div>
304 <div class="col-6 col-sm-4 col-md-3 col-lg-2">
305 <a href="category.php?category=camera" class="category-card d-block text-decoration-none">
306 
307 <h3>Camera</h3>
308 </a>
309 </div>
310 <div class="col-6 col-sm-4 col-md-3 col-lg-2">
311 <a href="category.php?category=mouse" class="category-card d-block text-decoration-none">
312 
313 <h3>Mouse</h3>
314 </a>
315 </div>
316 <div class="col-6 col-sm-4 col-md-3 col-lg-2">
```

Orders

```
orders.php X
orders.php > html > body
16 <html lang="en">
99 <body>
104 <div class="orders-section my-5">
105 <div class="orders-title"><i class="fas fa-box text-warning"></i> Placed Orders</div>
106 <div>
107 <?php
108     if($user_id == ''){
109         echo '<div class="empty"><i class="fas fa-sign-in-alt fa-2x mb-3"></i><br>Please login to see your orders</div>';
110     }else{
111         $select_orders = $conn->prepare(query: "SELECT * FROM `orders` WHERE user_id = ? ORDER BY id DESC");
112         $select_orders->execute(params: [$user_id]);
113         if($select_orders->rowCount() > 0){
114             while($fetch_orders = $select_orders->fetch(mode: PDO::FETCH_ASSOC)){
115                 ?>
116                 <div class="order-card mb-4">
117                     <p><i class="fas fa-calendar-alt me-2 text-warning"></i>Placed on: <span><?=$fetch_orders['placed_on']; ?></span></p>
118                     <p><i class="fas fa-user me-2 text-warning"></i>Name: <span><?=$fetch_orders['name']; ?></span></p>
119                     <p><i class="fas fa-envelope me-2 text-warning"></i>Email: <span><?=$fetch_orders['email']; ?></span></p>
120                     <p><i class="fas fa-phone me-2 text-warning"></i>Phone Number: <span><?=$fetch_orders['number']; ?></span></p>
121                     <p><i class="fas fa-map-marker-alt me-2 text-warning"></i>Address: <span><?=$fetch_orders['address']; ?></span></p>
122                     <p><i class="fas fa-credit-card me-2 text-warning"></i>Payment Method: <span><?=$fetch_orders['method']; ?></span></p>
123                     <p><i class="fas fa-shopping-basket me-2 text-warning"></i>Your Orders: <span><?=$fetch_orders['total_products']; ?></span></p>
124                     <p><i class="fas fa-money-bill-wave me-2 text-warning"></i>Total Price: <span>RS.<?=$fetch_orders['total_price']; ?></span></p>
125                     <p>
126                         <i class="fas fa-info-circle me-2 text-warning"></i>
127                         Payment Status:
128                         <span class="order-status <?=$fetch_orders['payment_status'] == 'pending' ? 'pending' : 'completed'; ?>">
129                             <?=$fetch_orders['payment_status']; ?>
130                         </span>
131                     </p>
132                 </div>
133             }
134         }
135     }
136 }
```

Shop

```
search_page.php shop.php X
shop.php > html > body
18 <html lang="en">
178 <body>
232 <!-- Bootstrap 5 JS -->
233 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
234 <!-- jQuery (for AJAX) -->
235 <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
236 <script>
237 $(function() {
238     // Add to Cart AJAX
239     $('.add-to-cart-btn').click(function() {
240         var btn = $(this);
241         var productId = btn.data('product-id');
242         var name = btn.data('product-name');
243         var price = btn.data('product-price');
244         var image = btn.data('product-image');
245         var qty = $('<input type="text" value="">').val() || 1;
246
247         $.ajax({
248             url: '', // same PHP page
249             method: 'POST',
250             data: {
251                 add_to_cart: true,
252                 pid: productId,
253                 name: name,
254                 price: price,
255                 image: image,
256                 qty: qty
257             },
258             success: function(response) {
259                 showMessage('Added to cart!');
260             }
261         });
262     });
263 }
```

Wishlist

```
wishlist.php
nl > body > div.wishlist-section.my-5 > div.row.g-4 > div.col-12.col-sm-6.col-md-4.col-lg-3 > form.product-card.h-100.position-relative > div.product-actions.mt-2 > button.btn.btn-delete
31 <html lang="en">
228 <body>
233 <div class="wishlist-section my-5">
234 <div class="wishlist-title"><i class="fas fa-heart text-warning"></i> Your Wishlist</div>
235 <div class="row g-4">
236 <?php
237 $grand_total = 0;
238 $select_wishlist = $conn->prepare(query: "SELECT * FROM `wishlist` WHERE user_id = ?");
239 $select_wishlist->execute(params: [$user_id]);
240 if($select_wishlist->rowCount() > 0){
241     while($fetch_wishlist = $select_wishlist->fetch(mode: PDO::FETCH_ASSOC)){
242         $grand_total += $fetch_wishlist['price'];
243     }
244 <div class="col-12 col-sm-6 col-md-4 col-lg-3">
245 <form action="" method="post" class="product-card h-100 position-relative">
246 <input type="hidden" name="pid" value="<?=$fetch_wishlist['pid']; ?>">
247 <input type="hidden" name="wishlist_id" value="<?=$fetch_wishlist['id']; ?>">
248 <input type="hidden" name="name" value="<?=$fetch_wishlist['name']; ?>">
249 <input type="hidden" name="price" value="<?=$fetch_wishlist['price']; ?>">
250 <input type="hidden" name="image" value="<?=$fetch_wishlist['image']; ?>">
251 <a href="quick_view.php?pid=<?=$fetch_wishlist['pid']; ?>" class="btn-view" title="Quick View"><i class="fas fa-eye"></i></a>
252 ">
253 <div class="name"><?=$fetch_wishlist['name']; ?></div>
254 <div class="price">RS.<?=$fetch_wishlist['price']; ?>/</div>
255 <input type="number" name="qty" class="qty-input" min="1" max="99" onkeypress="if(this.value.length == 2) return false;" value="1">
256 <div class="product-actions mt-2">
257 <button type="submit" class="btn btn-add-cart" name="add_to_cart"><i class="fas fa-cart-plus"></i> Add to Cart</button>
258 <button type="submit" class="btn btn-delete" name="delete" onclick="return confirm('Delete this from wishlist?');">
259 <i class="fas fa-trash"></i> Delete</button>
260 </div>
261 </form>
262 </div>
```

Update Products

```
wishlist.php  placed_orders.php  update_product.php x
admin > update_product.php > html > head > style > .btn-update-product
90 <html lang="en">
196 <body>
200 <div class="update-product-section my-5">
202     <?php
203         if(isset($message)){
204             foreach($message as $msg){
205                 echo <div class="alert alert-info alert-dismissible fade show m-3" role="alert" style="border-radius:2rem;">
206                     <span>'. $msg.'</span>
207                     <button type="button" class="btn-close" data-bs-dismiss="alert" aria-label="Close"></button>
208                 </div>;
209             }
210         }
211         $update_id = $_GET['update'];
212         $select_products = $conn->prepare(query: "SELECT * FROM `products` WHERE id = ?");
213         $select_products->execute(params: [$update_id]);
214         if($select_products->rowCount() > 0){
215             while($fetch_products = $select_products->fetch(mode: PDO::FETCH_ASSOC)){
216                 <?>
217                 <form action="" method="post" enctype="multipart/form-data" class="update-product-form">
218                     <input type="hidden" name="pid" value="<?= $fetch_products['id']; ?>">
219                     <input type="hidden" name="old_image_01" value="<?= $fetch_products['image_01']; ?>">
220                     <input type="hidden" name="old_image_02" value="<?= $fetch_products['image_02']; ?>">
221                     <input type="hidden" name="old_image_03" value="<?= $fetch_products['image_03']; ?>">
222
223                     <div class="product-image-preview mb-3">
224                         
225                         
226                         
227                     </div>
228
229                     <div class="mb-3">
230                         <label class="form-label">Product Name</label>
```

Chapter 7: Conclusion & Future Work

Summary

The purpose of the study was to develop an e-commerce website for Momal, Pakistan Occidental. The researchers used content management system to develop the website. Also, tools and plugins were utilized to make the website even more interactive. Base on the result of the testing and evaluation, the study was able to meet user requirements.

Conclusion

Based on the result of the testing, it was concluded that the system was able to perform all functions base on the objectives of the study. The result of the evaluation shows that the users agreed to use the system. It also showed that the task difficulty of using the system was easy.

Recommendations

During the testing of the system, the following recommendations were sought:

1. Add more modes of payments particularly the following:
 - a. Bayad Center
 - b. Debit Card
 - c. Bank Transfer
 - d. Electronic Bill Payment
 - e. Mobile Payments and
 - f. Electronic Check
2. Add an email notification feature that will inform the vendor that an order has been made by a customer.

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